

Tonogenesis in Northern Pomo
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QP 2

Abstract

This paper investigates the diachronic development of a High vs. Low level tone contrast in Northern Pomo, a dormant Indigenous language of Northern California. Although the formation of level tones is typically associated with the loss of segmental contrasts, this pathway for Northern Pomo's tones is not supported by comparative evidence. Instead, Northern Pomo's tones are argued to have developed through the reanalysis of a prominence marking accent and interactions with syllable structure. This reanalysis is motivated through a discussion of the effects that the segmental composition of syllables may have on the perception of pitch movement and height.

1. Introduction

Northern Pomo (ISO: pej), a dormant indigenous language of Northern California (O'Connor 1992), exhibits a lexical tone system in which level (H)igh and (L)ow tones¹ may be contrasted on heavy syllables e.g. *má:* 'you (pl.)' vs. *mà:* 'ground, earth; thing' and *t^hót* 'rotten' vs. *mòts* 'sour'. This contrast has not been reported in any of the six related Pomoan languages and comparative evidence suggests that Northern Pomo's tonal contrasts developed through three distinct and crucially ordered processes: (1) the association of a L tone to accented long vowels in Proto-Western Pomo, (2) the contraction of certain disyllabic sequences and the vocalization of certain word final glides creating new phonemically long vowels which had not developed L tone, and (3) the shortening of phonemically long vowels in closed syllables creating new short vowels without altering tonal associations and thus extending the H vs. L contrast to some syllables with short vowels.

In this paper, I argue that the association of L tone to accented long vowels proceeded through the reanalysis of the historical Pomoan Accent. This path for *tonogenesis* (the diachronic development of tonal contrasts; see e.g. Kingston 2011; Ratliff 2015; and Michaud and Sands 2020 for overviews) differs significantly from the commonly attested process of *segmental contrast neutralization* (the emergence of tonal contrasts co-occurring with the loss of segmental contrast such as voiced vs. voiceless in the syllable onset or in the syllable coda, glottal stop vs.

¹ Throughout this paper I will use "fundamental frequency" or "f0" to refer to the physical measurement of a vocal fold vibration, "pitch" to refer to its perceptual correlate and "tone" to refer to its phonological representation.

glottal fricative, or glottalic vs. non-glottalic consonant.) Segmental contrast neutralization is tied to phonetic mechanisms by which different consonantal contrasts lead to systematic and perceptible differences in f0 during adjacent vowels (Hombert 1978; Hombert, et al. 1979). However, these mechanisms appear to have played little to no role in the development of Northern Pomo's lexical tone. Instead, I ground the reanalysis of the historical Pomoan Accent in current research on the effects of segmental material on the perception of f0 movements (e.g. House 1990) and pitch-accent scaling (Barnes et al. 2024).

This paper proceeds as follows: Section 2 gives a brief overview of tone and tonogenesis. Section 3 gives an introduction to Northern Pomo, placing it within the Pomoan languages and giving an overview of its lexical tones. Section 4 outlines the correspondences of Northern Pomo's H and L tones, establishing the contexts for their development and demonstrating that they cannot be accounted for through segmental neutralization. Section 5 discusses the perceptual mechanisms that may motivate the reanalysis of the historical Pomoan Accent as a L tone on CVV(C) syllables. Section 6 concludes.

2. Tone and Tonogenesis

Many, if not most, languages around the world employ tone as a minimally contrastive feature distinguishing one word from another (Yip 2002, pp. 1, 17). In the prototypical case, a tone language contrasts numerous different tones, both level and contour, on every word or syllable (see Hyman 2009). Take for example the six way contrast in Hanoi Vietnamese as characterized by Kirby (2011, p. 386): *ma* [ma 1] 'ghost' with a level tone, *mà* [ma 1] 'but, yet' with a mid-falling tone, *má* [ma 1] 'cheek' with a rising tone, *mạ* [ma 1] 'rice seedling' with a low glottalized tone, *mả* [ma 1] 'tomb' with a low falling tone, and *mã* [ma 1] with a broken tone.²

2.1 Tonogenesis through the loss of consonantal contrasts

In many cases - Vietnamese being perhaps the best known example - these tonal contrasts can be traced back to a point of origin, to their *tonogenesis* (Kingston 2011; Ratliff 2015;

² Although these contrasts are realized in terms of complexes of phonetic features beyond f0, Brunelle (2009) presents perceptual and phonological evidence, arguing that they are better represented through abstract tonal categories rather than through bundles of phonetic features. This supports the use of Vietnamese as an expository example here, in line with its role as a standard example in the tonogenesis literature.

Michaud & Sands 2020).³ Tonogenesis refers to the development of tonal contrasts in a previously nontonal language. In the case of Vietnamese, the development of its rich set of tonal contrasts proceeded through at least two distinct phases shown schematically in Table 1 (Haudricourt 1954/2018; Pham 2003). In phase 1, the loss of syllable final **ʔ* and **h* coincided with the development of a rising contour and a falling contour respectively. Words without these final consonants developed a level mid tone in contrast to these contours. In phase 2, there was a merger of the voiced and voiceless obstruents in onset position which coincided with the splitting of the already established contours, e.g. the mid-level tone preceded by voiceless stops remained mid-level [1], but became mid-falling [ɿ] when preceded by voiced stops. This basic model of tonogenesis is taken to be prototypical and has been implicated in the development of tonal contrasts across Southeast Asia (Matisoff 1973) including various Sinitic, Tai-Kadai and Hmong-Mien languages (Michaud & Sands 2020).

Original		Phase 1		Phase 2	
pa ba	Non Tonal	pa 1 ba 1	Mid-Level	pa 1	Mid-Level
				pa ɿ	Mid-Falling
paʔ baʔ	Non Tonal	pa ɿ ba ɿ	Rising	pa ɿ	Rising
				pa ɿ	Low-Glottalized
pah bah	Non Tonal	pa ɿ ba ɿ	Falling	pa ɿ	Low-Falling
				pa ɿ	Broken

Table 1: Schematic representation of the development of Vietnamese's 6 way tonal contrast adapted from Pham (2003) citing Haudricourt (1954). In this schematic, *p* stands in for all voiceless obstruents and *b* stands in for all voiced obstruents.

Although the term *tone language* might invoke the (aural-)image of paradigmatic contrasts between a rich set of level and contour tones, it is common for languages to exhibit contrasts between only two or three tones. In these cases, tonogenesis is usually tied to one kind

³ According to Ratliff (2015) the term tonogenesis was first coined by James Matisoff (1970). However, research on the historical development of tonal contrasts dates back at least to Maspero's (1912) work on Annamese (the French colonial name for Vietnamese) historical phonetics.

of consonantal merger but not both. For instance, the development of tone through the loss of glottal contrasts in the syllable's coda has been observed in a variety of languages around the world such as Athabaskan (Kingston 2005; Krauss 2005) as well as a number of Mayan languages (see Bennett 2016, sec. 6.2 and references cited therein). In these cases, there was no subsequent merger in the syllable onset to further split the contrasts. Additionally, the loss of contrasts in the onset have been widely attested throughout southeast asia without a preceding merger in the coda (Matisoff 1973). Well known examples are the Kammu⁴ (Svantesson 1989; Svantesson & House 2006) languages and Blang (Svantesson 1989) of the Mon-Khmer family, as well as, the Tibeto-Burman languages Lahu (Matisoff 1970; Mazaudon 1977) and Karen (Mazaudon 1977) among others (see Gehrman 2022 for a comprehensive survey of tonogenetic patterns in Southeast Asia).

2.1.1 Phonetic ties between segmental contrasts and f₀

Researchers argue that tone develops through the loss of segmental contrasts because the vocal fold configurations during the articulation of these contrasts induce systematic differences in the f₀ of neighboring vowels (Lehiste 1970; Halle & Stevenson 1971; Ohala 1978; Honda et al. 1999; cf. Kingston & Diehl 1994). Hombert (1978) validates the plausibility of f₀ perturbations induced by glottal segments in coda position by presenting a range of studies. In a production experiment with 4 Arabic speakers, Hombert (1978:92-94) shows that f₀ rises about 9Hz in vowels immediately preceding a glottal stop -ʔ but falls about 25Hz before a glottal fricative -h. Following this, a perceptual experiment with 10 American English speakers demonstrates that even speakers of a nontonal language are able to perceive f₀ differences akin to those induced by coda -ʔ and -h in Arabic (Hombert 1978:94-95). The fact that segmentally induced perturbations can be both produced and perceived by speakers of nontonal languages like English and Arabic is taken as evidence of their role in tonogenesis through segmental contrast neutralization (see also Hombert et al. 1979).

Although onset voicing is often considered to be responsible for splitting already existing tones (e.g. Kingston 2011), the relationship between onset voicing and f₀ perturbations has seen much more attention than the relationship between f₀ and glottal codas. Early work (e.g. House & Fairbanks 1953; Lehiste & Peterson 1961; and Mohr 1971) reports that the average f₀ in

⁴ An alternative spelling found in the literature is *Khmu*

vowels following a voiced stop is systematically lower than the average f0 of a vowel following a voiceless stop. This is corroborated by a string of experiments reported by Hombert (1978) and Hombert, et al. (1979) replicating the effect in both tonal (e.g. Yoruba) as well as nontonal (e.g. English and French)⁵ languages.

Moreover, Hombert (1978) and Hombert et al. (1979) show that the f0 perturbations on the order of those with the voicing contrasts are audible to speakers of American English, a nontonal language, through a series of perceptual experiments using stimuli intended to simulate f0 perturbations analogous to those associated with the voicing contrast.

2.2 Tonogenesis through vowel height and length

In isolated cases, tonal contrasts have been introduced through changes in the vowel system. For example, in the Angkuic languages⁶ U (Svantesson 1988, 1989) and Hu (Svantesson 1989, 1991) a phonemic vowel length contrast developed into a H vs. L tonal contrast with H tone on historically short vowels and L tone on historically long vowels. That the development of tone in Hu was triggered by phonemic vowel length and not onset voicing can be demonstrated by comparing Hu forms to their Northern Kammu and Southern Kammu cognates.^{7,8}

Southern Kammu has not undergone tonogenesis and retains both a voicing contrast in syllable onsets and a phonemic vowel length contrast. Northern Kammu has lost the onset voicing contrast with H tone developing after historically voiceless consonants and L tone developing after historically voiced consonants. For example the Northern Kammu word *táaŋ* ‘to weave’ developed a H tone after the historically voiceless *t* (cf. Southern Kammu *taaŋ* ‘to weave’) while the Hu cognate *tʰàŋ* ‘to weave’ developed a L tone in the same context. The development of L tone here is concomitant with the loss of phonemic vowel length. Similarly, Northern Kammy *priʔ* ‘forest’ developed a L tone after an onset with a historically voiced stop

⁵ The pair of languages, English and French, in these experiments also shows that these differences in f0 are born out in languages where the voicing contrast is realized phonetically as voiceless unaspirated vs. voiced (e.g. French) as well as in languages where it is realized as voiceless aspirated vs. voiceless unaspirated in word initial positions (e.g. English).

⁶ The Angkuic languages are a group of Mon-Khmer languages spoken in Yunnan province, China (Hsiu 2015).

⁷ As mentioned in Section 2.1, Northern Kammu, among other Kammu dialects, developed tonal contrasts through a merger in onset voicing. As the discussion here suggests, the tonal developments in the Anguik languages Hu and U are independent from those in the Kammu dialects. See Svantesson (1989) for a full account of the different outcomes.

⁸ The Hu and Kammu examples cited here are adapted from Svantesson’s (1989) Table 2, Table 8 and Table 9. Svantesson (1989) does not provide reconstructions, so I present Southern Kammu forms as a proxy

(cf. Southern Kammu *bri?* ‘forest’) while Hu *pɛi?* ‘forest’ developed a H tone in the same context, reflecting instead the historical short vowel.

There are also instances in which features of a vowel’s quality (e.g. $\pm$ high or $\pm$ Advanced Tongue Root) give rise to tonal contrasts. Examples of these changes are U (Svantesson 1989), Lugbara (Andersen 1986) and Mangbetu (Demolin 1999, as cited in Michaud & Sands 2020). However, in each case, the role of vowel quality is to split or otherwise change already existing tones. There are no known cases in which vowel quality is responsible for introducing tonal contrasts into an otherwise nontonal language.

2.2.1 Phonetic ties between vowel contrasts and tonogenesis

The relationship between vowel height and f0 is well established and is often assumed to be a universal consequence of vowel articulation referred to as *vowel intrinsic f0 effects*. In every language (both tonal and nontonal) for which these effects have been investigated, high vowels are shown to have higher f0 than low vowels (see Whalen & Levitt 1995 for a meta-analysis of 62 studies on 31 different languages). As with the consonantal effects on f0 discussed above, the intrinsic f0 effects of vowels appear to be tied to articulatory mechanisms. Essentially, movements of the tongue and jaw may pull the vocal folds stiff in high vowels and leave them slack in low vowels (Chen et al. 2021). The fact that the intrinsic f0 effects of vowels so rarely participate in the development of tone is surprising given its apparent universality. This may be due to the fact that listeners appear to compensate to some degree for the effects of vowel quality in the perception of pitch (Chuang & Wang 1978; Hombert 1978; Stoll 1984). This capacity might make the f0 differences that are intrinsic to vowel quality less likely to be phonologized than those incurred by consonants.⁹

Unlike vowel quality, phonemic vowel length has led to the development of tone in at least one language. This is particularly surprising given that there is not a clear phonetic mechanism that would lead to higher f0 on short vowels and lower f0 on long vowels. In the case of Hu, Svantesson (1989) suggests that the development of L on long vowels may be related to the tendency for f0 to decline over the course of an utterance. While vowel quality contrasts may affect f0 through shifts in tongue and jaw positions, the position of the tongue and jaw might be

⁹ Hombert et al. (1979) also suggest that the fact that f0 perturbations caused by consonants are realized in the neighboring vowel and as such are separable (perceptually) from their source may also play a role in this asymmetry. The intrinsic f0 effects of vowels are realized within their source and thus are harder to separate from the vowel.

expected to remain more or less constant in a contrast between short and long vowels (Connell 2002, p. 111). To my knowledge, there is no systematic investigation of this relationship in a nontonal language.¹⁰ Among tonal languages, there are investigations looking into potential relationships between tonal categories and the duration of the vowels that they are produced in. One example is Gandour's (1977) investigation of Thai, which reports that vowels with H tone in Thai are systematically shorter than vowels with L tone. However, these findings are discussed in terms of the effect that tone has on vowel duration; it is unclear how well they apply to the tonogenetic potential of phonemic vowel length, which would depend on an effect in the inverse direction: vowel length would have to determine f_0 . In one study designed to test the effect in this direction, Connell (2002) reports that there was no significant effect of phonemic vowel length on f_0 at any tone height in Kunama, a Nilo-Saharan language with a 3-level tone system.

The difficulty in establishing an articulatory or acoustic link between vowel duration and f_0 paints vowel length as an unlikely source for the development of tone. Furthermore, Yu (2010) and Yu, Lee & Lee (2014) demonstrate that speakers of American English perceive syllables with higher f_0 to be longer than those with lower f_0 . This tendency is also reported in non-linguistic tasks, as demonstrated by Brigner (1988) who found that listeners perceive pure tones with high f_0 to be longer than pure tones with low f_0 despite the fact that each pure tone was identical in duration. The relationship between vowel duration and f_0 height is rather tenuous and may be readily compensated for in perception by speakers of at least some languages. Even if an effect of vowel length on f_0 were observed, it appears that listeners can readily compensate for it. As such, it is perhaps surprising that Hu underwent a change from phonemic vowel length to tone.¹¹

2.4 Tonogenesis through prosodic changes

Finally, some languages are considered to have developed tonal contrasts through changes in word and syllable structure. Perhaps the most well-known example of this kind of tonogenesis is the development of accentual contrasts in Swedish (North Germanic).¹²

¹⁰ House & Fairbanks (1953) and Mohr (1971) do report a tendency in English for shorter vowels to have higher f_0 than longer vowels. However, this correlation is likely driven by vowel quality. High vowels tend to both be shorter and have higher f_0 than low vowels.

¹¹ Indeed, Hu's abnormal path to tonogenesis might have been partially paved by the highly tonal areal pressures of Southeast Asia.

¹² Although modern Swedish exhibits pitch based lexical contrasts, the contrasts are not realized in terms of a paradigmatic opposition between multiple levels or contours as in prototypical tone languages. There is ongoing

Depending on the dialect of modern Swedish, the accentual distinction is characterized as a contrast between words with early (Accent 1) vs. late (Accent 2) pitch peaks relative to the stressed syllable, or a contrast between words with one (Accent 1) vs. two (Accent 2) pitch peaks. Examples include ¹gæster ‘guest’ (Accent 1) vs. ²døøma ‘to judge’ (Accent 2). The accent type is marked by the superscript number preceding the stressed syllable, and stressed syllables are indicated through underlining.

Although accounts differ in the timing and precise mechanisms involved, it is generally believed that this contrast developed through the lexicalization of a focus marking tone following changes to word or metrical structure (see Iosad 2024 for a concise overview). The basic insight, shared by all accounts, is that words which are attested with Accent 1 in the modern dialects correspond to monosyllabic words in Old Norse or are reconstructible with only one stressed syllable in Proto-Norse,¹³ while words attested with Accent 2 correspond to polysyllabic words in Old Norse or are reconstructed with multiple stresses in Proto-Norse. For example, ¹gæster ‘guest’ with Accent 1 in the modern dialects derives from monosyllabic Old Norse gæstr (reconstructed to Proto-Norse *gestiz) while ²døøma ‘to judge’ with Accent 2 in the modern dialects derives from disyllabic Old Norse døøma (reconstructed to Proto-Norse *døømi jan).

Two main hypotheses explain how these historical differences gave rise to the modern contrast. Riad (1998) traces the development to Proto-Norse, proposing that focus was marked by assigning a H tone to each stressed syllable. Words like *gestiz, with one stressed syllable, received one H tone and were realized with one pitch peak under focus, while words like *døømi jan, with two stressed syllables, received two H tones and were realized with two pitch peaks under focus. Syncope created stress clashes in words which historically had two stressed syllables, e.g. Proto-Norse *døømi jan > *døø man. The resulting stress clash would have synchronically been resolved by de-stressing the second syllable *døø man → [døøman], but both syllables would continue to receive H tones, maintaining two pitch peaks under focus despite having only one surface stress. As the number of pitch peaks under focus was no longer

debate as to whether the languages discussed in this section should be considered tone languages or if they should be treated as a separate category, often called “lexical pitch-accent” (see Hyman 2009, cf. van der Hulst 2011, *i.a.*). As a consequence, these languages are often left out of discussions of tonogenesis, an example being Ratliff (2015). On the other hand, Kingston (2011) discusses the case of Swedish as a case of phonetically based sound change, treating it identically to the cases of segmental neutralization discussed above.

¹³ Old Norse refers to the written language of the Viking age and medieval period (800 - 1350 CE) while Proto-Norse refers to the largely reconstructed language which is sparsely attested in runic inscriptions dating between 200 - 800 CE (Haugen 1976).

predictable from the number of stresses, the differences between words which received two and words which received one pitch peak was reinterpreted as being an aspect of their lexical representation rather than being a predictable part of focus marking.

Alternatively, Elstad (1980), Bye (2004, 2011) and Kristoffersen (2021) trace the development to Old Norse, proposing that all words had a single pitch peak under focus. In polysyllabic words (e.g. Old Norse '*døoma* 'to judge) the pitch peak could be delayed and realized in the syllables following the stressed syllable; however, in monosyllabic words (e.g. Old Norse '*gæstr* 'guest') it would have to be realized within the stressed syllable. Over time, changes in word structure, like epenthesis, altered syllable count without changing the alignment of the pitch peak: Old Norse '*gæstr* 'guest' > '*gæster* retains early alignment. Speakers then reinterpreted these alignment differences as being lexical. Both accounts share the crucial insight that accentual contrast formed as predictable variation in the realization of focus marking became opaque through structural changes, leading to the lexicalization of the tonal contrast.

2.4.1 Mechanisms for tonogenesis through prosodic change

In the case of tonogenesis through prosodic change discussed above, the development of tonal contrasts are spurred by alterations to word and syllable structures, or changes in prominence relations. Unlike segmental contrast neutralization where tones develop out of sublexical phonetic cues to lexical contrasts, tonogenesis through prosodic change involves the reanalysis of an already established intonational tone from the phrasal level into the lexical representation of individual words. For Riad (1998) in Proto-Norse each stressed syllable in a focused word was marked with a H tone, leading to a pitch peak on every stressed syllable in the word. For Elstad (1980), Bye (2004, 2011), and Kristoffersen (2021) focus marking was always realized with a single pitch peak on the stressed syllable. However, in Old Norse the pitch peak was realized later in polysyllabic words than it was in monosyllabic words. In either case, sound changes altering the structure of words eliminated the contexts that originally conditioned predictable differences in the realization of focus marking and led speakers to integrate tonal information that was once provided by the focus marking system into the lexical representation of individual words.

While tonogenesis through segmental contrast neutralization is motivated by the articulatory and perceptual link between f_0 and the realization of certain segmental contrasts

(Ohala 1981; Kingston & Diehl 1994; Gao & Kirby 2024), no phonetic, articulatory or perceptual link has been established between f_0 and changes to word, syllable or metrical structure. Instead, tonogenesis through prosodic change - at least in the case of the Swedish tonal accents - appears to be motivated by the creation of opacity in the once predictable association and realization of intonational tones within individual words. This opacity leads the speaker to reanalyze an intonational or phrasal tone - already present in the language's phonology - into the representation of certain lexical items. (This may be viewed as a kind of domain generalization in the sense of Myers & Padgett (2014).) Finally, it is worth noting, that the resulting contrasts are not realized in terms of discrete differences in pitch levels or contours and they do not seem to be represented in terms of a paradigmatic set of contrasting tone categories associated with individual syllables as is the case in the prototypical tone languages discussed above.

2.5 Summary

Tonogenesis is a widely attested phenomenon that has occurred in the history of languages around the world. In this section, I outlined two general pathways. The prototypical pathway to tonogenesis, segmental contrast neutralization, is grounded in phonetic and perceptual mechanisms: certain segmental contrasts (e.g. onset voicing, coda glottalization, vowel quality or vowel length) are associated with “mechanical differences in pitch” (Hombert, et al. 1979). Over time, physiologically conditioned differences in pitch become tones as they supplant the segmental contrasts which originally caused them (Ohala 1981, 1989, 1993; Gao & Kirby 2024).

In cases where changes to the prosodic structure lead to the development of lexical tone contrasts, intonational tones are reanalyzed as part of the lexical representation of individual words. Unlike segmental contrast neutralization, tonogenesis that occurs through prosodic changes are not obviously driven by phonetic and perceptual mechanisms. Instead, the reanalysis of an intonational tone is driven by opacity in its realization that is created by changes in prosodic structure.

These two pathways often lead to drastically different kinds of tonal contrasts as well. The tone systems that develop through segmental contrast neutralization may be characterized by rich paradigmatic contrasts between level and/or contour tones. On the other hand, the tone systems that develop through changes in prosodic structure are often syntagmatic in nature,

contrasting either in the number or alignment of pitch movements within a word. In this respect, Northern Pomo's tonogenesis presents a challenge. As I will argue in Section 4, Northern Pomo's tonal contrasts are tied to the reanalysis of a prominence marking "accent" spurred on by alterations to word and syllable structure. While the expected outcome might be a contrast between early vs. late aligned pitch movements or different numbers of pitch peaks in words with similar structures (à la Swedish), Northern Pomo's contrast is realized in terms of H vs. L pitch levels. Without evidence of segmental contrast neutralization, there is no clear model for the development of a level tone contrast, an issue which I turn to in Section 5. But first, I will provide an introduction to Northern Pomo, situate it within the Pomoan language family and present an overview of its system of lexical tones.

3. Northern Pomo, tone and accent in the Pomoan language family

3.1 The Pomoan language family

The Pomoan language family is composed of seven distinct languages: Southeastern, Eastern, Northeastern, Northern, Kashaya, Southern and Central Pomo. These languages are indigenous to Northern California and were spoken in what are Mendocino, Sonoma, Lake and Colusa Counties today. The Pomoan languages exhibit a range of linguistic diversity comparable to the Germanic languages (McLendon and Oswalt 1978) making the languages within this family well suited for comparative and historical study.

There have been a number of proposals concerning the internal relationships of the Pomoan languages.¹⁴ In this paper I will follow the classification proposed by Oswalt (1976; see Figure 1). In this model, the Western Branch, composed of Northern, Kashaya, Central and Southern Pomo constitute a genetic subgrouping distinct from the other three Pomoan languages: Southeastern, Eastern, and Northeastern Pomo, which form individual branches of the Pomoan language family. Southeastern Pomo was the first language to diverge from the Pomoan group, followed by Eastern with both settling along the shores of Clear Lake. Northeastern Pomo diverged with its speakers moving into the Sacramento valley as the speakers of the Western Branch moved into the Russian River drainage. Northern Pomo was the first to diverge from the

¹⁴ Alternative phylogenies have been proposed by Kroeber (1925, p. 227), Halpern (1964), Oswalt (1964), Webb (1971), and Halpern (1984), . The phylogenetic tree proposed by Kroeber (1925, p. 227) is based on a lexicostatistical classification made by Barrett (1908). Each of these classifications is provided in McLendon (1973, pp. 3-5) with the exception of Halpern's (1984) classification and Buckley (2024) provides a detailed, yet concise summary of the similarities and differences between a number of the most relevant phylogenetic proposals.

rest of the Western Pomo languages, as the speakers of Kashaya, Southern, and Central Pomo migrated southward along the Russian River and its tributaries forming the Western Branch's Southern Group.¹⁵

Throughout this paper I will refer to the Northern, Kashaya, Southern and Central Pomo collectively as the Western Pomo languages and Northeastern, Eastern and Southeastern Pomo as the non-Western Pomo languages. It should be emphasized that the designation *non-Western Pomo languages* is not meant to suggest that Southeastern, Eastern and Northeastern Pomo form a distinct genetic subgrouping, it is only intended as a term of convenience indicating that these languages are excluded from the Western Branch. The reconstructed language of the whole family will be called Proto-Pomo, the reconstructed language of the Western Pomo languages will be referred to as Proto-Western Pomo.

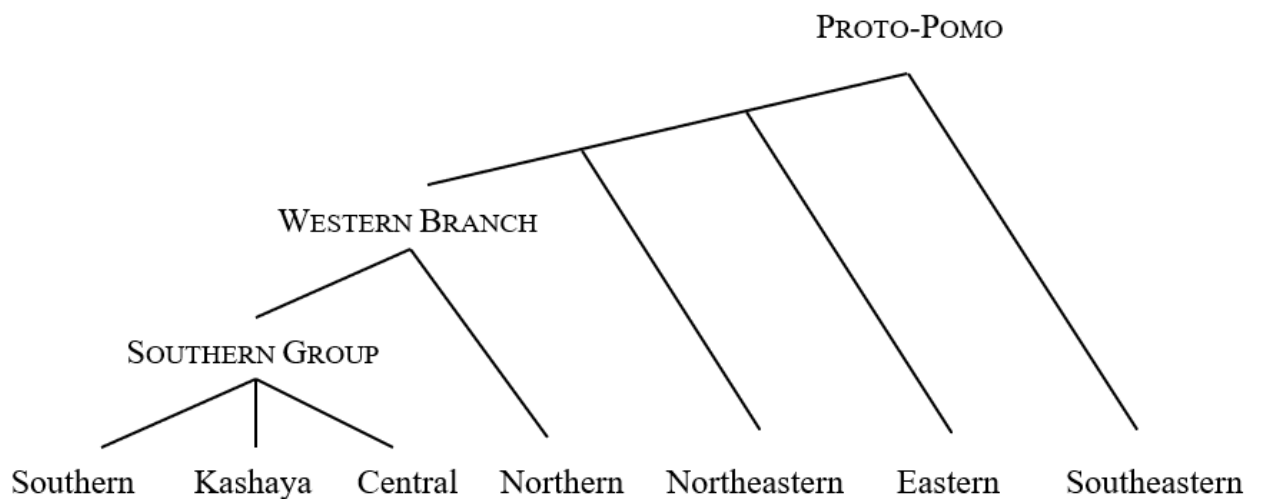


Figure 1: Adaptation of Oswalt's (1976) genetic classification of the Pomoan family to appear in Buckley (2024) and provided to the author by Buckley (p.c.).

3.2 Northern Pomo

Northern Pomo was spoken largely within what is today Mendocino County in an area extending from the northwest shore of Clear Lake to the Pacific Coast (Barrett 1908, p. 125-127). Although the last fluent native speaker of Northern Pomo passed away in 2005,

¹⁵ For a discussion of the potential time depths and geographical patterns of the Pomoan settlements, divergences and migrations, see Whistler (1983).

extensive field work throughout the mid to late 20th century has produced an audio-corpus containing approximately 900 hours of recordings. This corpus is composed of conversations, stories and narratives, as well as word-lists and targeted linguistic elicitations from the three last known living speakers: Edna Campbell Guerrero, Annie Lake and Elenor Stevenson Gonzales. The recordings were collected by Eero Vihman, working with Edna Campbell Guerrero and Annie Lake in the 1960s, and Cathy O'Connor, working with Edna and Annie, as well as Elenor Stevenson Gonzales from 1979 to 2005. This audio corpus forms the basis of an ongoing language documentation effort led by O'Connor. The current paper, as an extension of these efforts, draws extensively on the materials held within the corpus.¹⁶

Thus far, very little has been published concerning Northern Pomo's phonological system. O'Connor (1992, pp.8-9) provides a sketch of Northern Pomo's phoneme inventory. This includes 29 consonants, 5 distinct vowel qualities and a binary distinction in vowel length (i.e. short vs. long).¹⁷ Northern Pomo's syllable structure is fairly restricted, disallowing onset clusters and allowing coda clusters only in specific morphological contexts. Additionally, phonemically long vowels are disallowed in closed syllables and are realized phonetically as a short vowel when a morphological context would otherwise derive a CVVC syllable; this process will be discussed further in Section 3.4.2.1.

The traditional description of Northern Pomo's phonological system includes a phonemic or lexical pitch-accent, typically on either the first or second syllable of a word (McLendon 1973; Vihman 1976).¹⁸ However, subsequent research suggests that the traditional description of Northern Pomo's "lexical pitch-accent" conflates word-level metrical prominence (i.e. "stress"; Dailey 2023) and predictable f0 movements (Dailey 2020).¹⁹ Additionally, the presence of lexical

¹⁶ In addition to the audio recordings contained in this corpus, fieldwork was carried out by Abe Halpern and Robert Oswalt. Little of the work has been published, although handwritten field notes are available at the California Language Archives. Thus far, these materials have not been integrated into this project.

¹⁷ Eero Vihman, at some point, suggested that Northern Pomo had a ternary length distinction, contrasting short, long and extra-long vowels (O'Connor p.c.) This may be related to his L1 Estonian which is reported to have a ternary vowel length distinction in which a salient perceptual cue to the "overlong" category is a distinctive pitch movement (Lehiste 1975).

¹⁸ Vihman's full account split syllables into 4 categories: "short rising" composed of CV syllables and syllables closed with a non-glottalized consonant; "short falling" composed of syllables closed with ejective consonants and glottal stops; "long rising" composed of syllables closed with nasals, glides and the lateral sonorant; and "long falling" composed of syllables with long vowels. Essentially, if the accent is on a syllable with a phonemically long vowel (CVV) syllable or one that is closed by a glottalic consonant (ejectives and ?) then it is produced with a low or falling f0 contour, but if it falls on a light syllable (CV) or a syllable closed with any other consonant, then it is produced with a high or rising f0 contour.

¹⁹ These findings are consistent with recent advances within the field of word-level prosodic typology where the coherence of "lexical pitch-accent" as a language type has been called into question (Hyman 2009). Instead,

tone has been suggested (Halpern 1984, fn. 36; Shepardson 2008)²⁰ but has not yet been fully described. The remainder of this section is intended to establish the presence of lexical tone in Northern Pomo and show its interactions with a predictable f0 movement.

3.3 Word-level metrical structure, predictable f0 movements, and long vowels

The traditional description of “lexical pitch-accent” asserts that word-level prominence in Northern Pomo is primarily expressed through distinctive f0 movements associated with either the first or second syllable of the word (McLendon 1973; Vihman 1976). Shepardson (2008) presents evidence from monosyllabic and disyllabic words suggesting that accented CV syllables are associated with a high or rising f0 while accented CVV syllables are realized with a low or falling f0. This pattern is shown below with the words *fa* ‘fish’ and *fà:* ‘arm’ in Figure 2 and *finà* ‘boat’ and *finà:* ‘head’ in Figure 3.^{21, 22}

Figure 2 shows two monosyllabic words: *fa* ‘fish’ in the left panel and *fà:* ‘arm’ in the right panel. Both words were spoken by Elenor Stevenson Gonzales and are displayed with their pitch tracks in a range from 125 Hz to 250 Hz so that f0 may be visually compared across panels. This pair contrasts only in phonemic vowel length. In *fa* ‘fish’ the vowel has a duration of 123 ms and an average f0 of 198 Hz and a slightly rising contour shown by the pitch tracker. In *fà:* ‘arm’, the vowel is nearly twice as long at 243 ms while the average f0 is much lower at about 156 Hz with a falling f0 contour shown by the pitch tracker.

descriptions of such systems may be shown to conflate phrasal with word-level prosody (Gordon 2014) or they may be better analyzed as lexical tone systems in which tonal contrasts are restricted by metrical or syllable structure (Hyman 2009).

²⁰ Halpern’s work on Northern Pomo is based on data that he collected in 1939 and 1940. He was the first to notice a correlation between phonemically long vowels and low pitch in Northern Pomo, mentioning it in a talk given at Berkeley in 1962 and the first to point out the presence of the minimal pair *má:* ‘you pl.’ vs. *mà:* ‘earth, ground; things’ (Halpern 1984, fn 36) as well as other words with phonemically long vowels and high pitch: *bitá:* ‘bear’ and *bá:* ‘person’. However, Halpern follows the tradition of Pomo scholarship in discussing the distinctive pitch as reflexes of Northern Pomo’s lexical “accent”, not lexical tone.

Shepardson (2008) also notes the phonemically long vowel with a high pitch in *bitá:* ‘bear’ and points out a previously unnoticed tonal difference in the determiners *nàm* ‘the’ vs. *náj* ‘that’. These determiners are not discussed in depth in this paper. They appear to have developed after Northern Pomo diverged from the rest of the Western Pomo languages and will require internal reconstruction work.

²¹ The representative examples of *fa* ‘fish’ and *fà:* ‘arm’ in Figure 2 as well as the representative examples of *finà* ‘boat’ and *finà:* ‘arm’ were produced by Elenor Stevenson Gonzales and found in the Talking Dictionary on northernpomolangaugetools.com. These were likely produced in isolation, however I have not yet been able to track down the sound file containing their original elicitation in order to confirm this.

²² The figures provided in this section were made by me in Praat (Boersma & Weenink 2025) using a script adapted from one written by José Joaquín Atria and posted on stack exchange on November 15th, 2016. In some cases, the annotations were formatted separately in Google Slides.

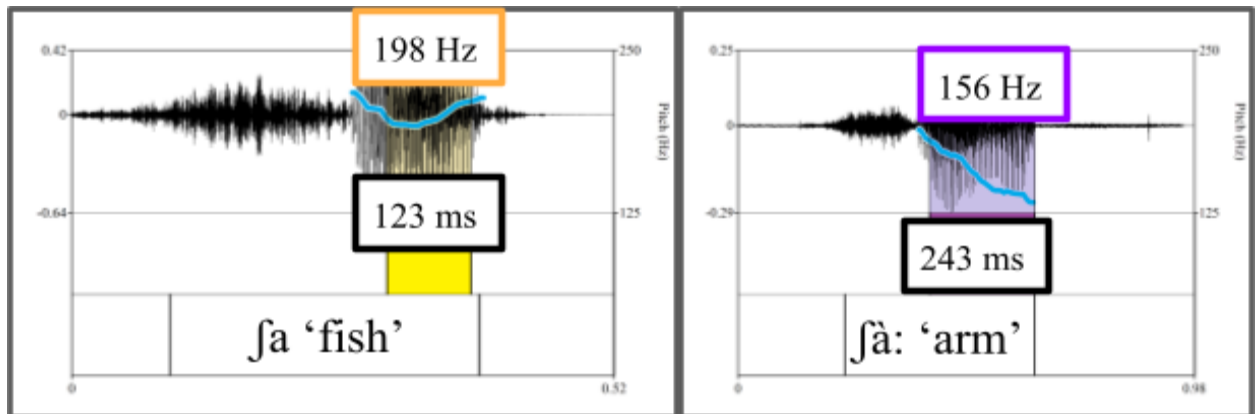


Figure 2: Pitch track and waveform for /ʃa/ ‘fish’ (left) and /ʃä:/ ‘arm’ (right). Both words were produced by Elenor Stevenson Gonzales.

The disyllabic near minimal pair /ʃina/ ‘boat’ and /ʃinä:/ ‘head’ is shown below in Figure 3. Again, both words are spoken by Elenor Stevenson Gonzales and are displayed with their pitch track in a range from 125 Hz to 250 Hz so that f0 can be visually compared across panels. In /ʃina/ ‘boat’ the vowel in the first syllable is about 66 ms in duration with an average f0 of 187 Hz while the vowel in the second syllable is 121 ms in duration with an average f0 of 195 Hz. The sharp fall in f0 on the first syllable shown by the pitch track may be an artifact of the voiceless fricative in its onset. Otherwise, the pitch track shows a rising f0 contour from the first syllable to the end of the second syllable. Following Dailey (2023), the increased duration of the second syllable may be tied to metrical prominence which happens to coincide with the location of the f0 peak.

In /ʃinä:/ ‘head’, the vowel in the first syllable is about 77 ms in duration with an average f0 of 160 Hz while the second syllable is 236 ms in duration with an average f0 of 132 Hz. The pitch track shows a generally falling f0 contour from the first syllable to the end of the second syllable. The phonemic vowel length contrast is demonstrated by the difference in duration between the final syllables of /ʃina/ ‘boat’ and /ʃinä:/ ‘head’ and as with the pair of words /ʃa/ ‘fish’ and /ʃä:/ ‘arm’ shown in Figure 3, the word composed of CV syllables is characterized by a high or rising contour while the word with a CVV syllable is characterized by a low or falling f0 contour.

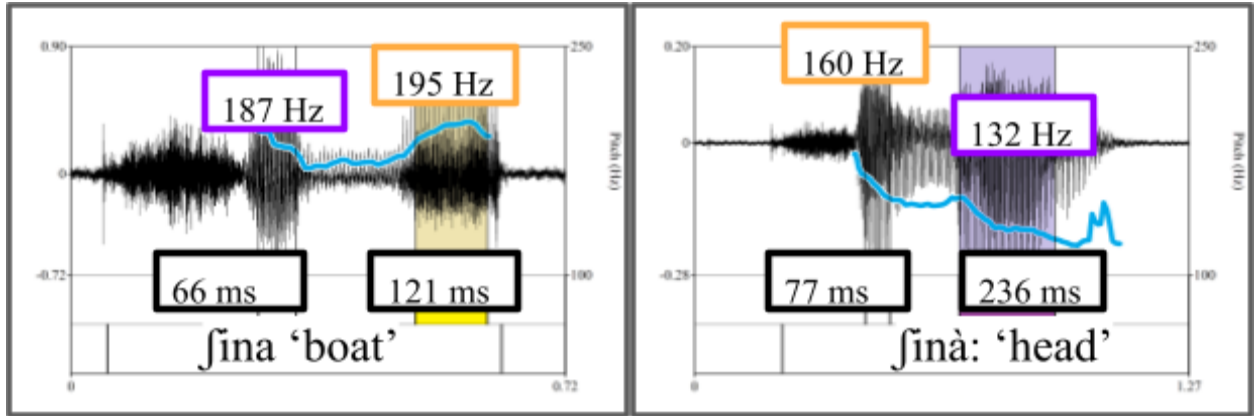


Figure 3: Pitch track and waveform for *fīna* 'boat' (left) and *fīnà:* 'head' (right). Both words were produced by Elenor Stevenson Gonzales.

Figure 4 shows the trisyllabic word *fadodo* 'wood owl' with three CV syllables in comparison to the word *fakò:do* 'rabbit' with a CVV in the second syllable.²³ Again, both words were spoken by the same speaker with f0 plotted in a range from 125 to 250 so that the pitch tracks may be visually compared across panels.

The vowel in the first syllable of *fadodo* 'wood owl' is about 95 ms duration with an average f0 of 160 Hz; the vowel in the second syllable is the longest in the word at about 166 ms in duration with the average f0 rising to 174 Hz and the vowel in the third syllable is about 126 ms in duration with the highest average f0 at 190 Hz. The pitch track shows a generally rising f0 contour with a slight dip in the first syllable, followed by a rising f0 through the second syllable and a high level f0 in the final syllable. The increased vowel duration could suggest that there is metrical prominence on the second syllable.

On the other hand, in *fakò:do* 'rabbit' the vowel in the first syllable is about 100 ms in duration with an average f0 of 173 Hz; the vowel in the second syllable is about 261 ms in duration with the average f0 falling to 166 Hz; and the vowel in the third syllable is about 77 ms in duration with the average f0 staying relatively level at 163 Hz. The vowel in the second

²³ The example for *fadodo* 'wood owl' was produced in the phrase *fadodonàm* in an elicitation led by Cathy O'Connor on July 10th 2001. The example for *fakò:do* 'rabbit' was found in the Talking Dictionary on *northernpomolangaugetools.com* and was likely produced in isolation. The increased duration of the second syllable may be contributed by a number of sources beyond metrical structure, including domain final lengthening. However, Dailey (2023) reports on a statistical analysis of syllable and segment duration which finds lengthening of the second syllable to be a general pattern in words composed of light syllables and is observed in words with more than two syllables, as well as words in different phrasal positions. The tokens discussed here are intended only as demonstrative examples of this general pattern.

syllable of *fakò:do* ‘rabbit’ is phonemically long at one and a half times the duration of the phonemically short vowel in the second syllable of *fadodo* ‘wood owl’. The pitch track shows a sharp fall in f_0 from the first syllable into the second syllable where it levels off and remains low through the final syllable.

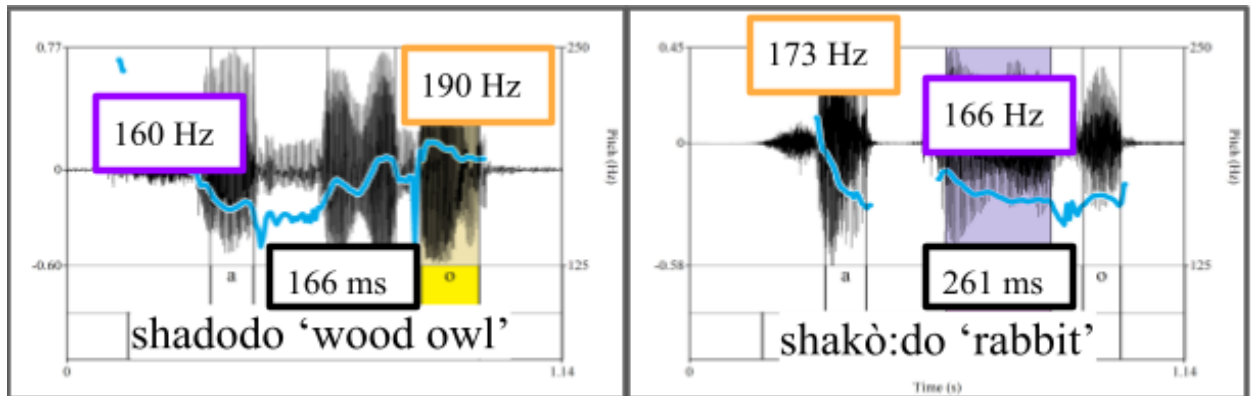


Figure 4: Pitch track and waveform for *fadodo* ‘wood owl’ (left) and *fakò:do* ‘rabbit’ (right). Both words were produced by Elenor Stevenson Gonzales.

The examples above show that words composed of CV syllables have a generally rising f_0 contour with a peak on the final syllable of the word, even when that syllable is not metrically prominent, as is seen in *fadodo* ‘wood owl’.²⁴ This suggests that the f_0 peak is marking the word’s edge instead of the metrically-prominent second syllable. However, Dailey (2020) cites words with internal closed syllables, such as *biṭúmṭu* ‘ant’ [bi.ṭúm.ṭu], *tsitoktok* ‘robin’ [tsi.ṭók.tok] and *Ṗíjtoktok* ‘butterfly’ [Ṗúj.tok.tok] where the f_0 peak is realized on the left-most closed syllable in each word, relating the distribution of f_0 peaks to a pattern of *left-to-right default-to-opposite* prominence assignment (Prince 1983; Gordon 2000). In other words, the left-most closed syllable receives a pitch peak, or the right most does if there are no closed

²⁴ This account of the distribution of f_0 peaks in Northern Pomo is a simplification. In words longer than 3 syllables, the f_0 peak does not always surface on the final syllable. In *misaxala* ‘snake’, the only monomorphemic word longer than 3 syllables that I have been able to find, there is some variability in the location of the f_0 peak. While the f_0 peak is often realized on the fourth syllable as is expected if it is assigned by an edge marking system, in some cases it is realized on the third syllable, with a high plateau or a slight fall through the fourth syllable. In the two examples that I have found where *misaxala* ‘snake’ is modified by an adjective, the f_0 peak is realized on the third syllable with a slight fall to through the fourth syllable with another f_0 peak on the final syllable of the adjective e.g. *misaxala bitʰu?* [misaxála bitʰúʔ] ‘small snake’. This leaves open the possibility that the f_0 peak is targeting the third syllable and not necessarily the final syllable in the word. Interestingly, if Northern Pomo is marking the third syllable with a pitch peak, this would bear a striking similarity to the system which assigns a “raised pitch” phrasal accent to the third syllable of sufficiently long words in the sister language Kashaya (Oswalt 1961, 1988; Buckley 1994b, 2023). However, more work is necessary before anything definitive can be said about this in Northern Pomo.

syllables. However, this pattern is interrupted with a phonemically long vowel, as the CVV syllable appears to be the target of low or falling f0 contour (e.g. *ʃakò:do* ‘rabbit’, Figure 4). At the present it is unclear which level of prosodic structure the pitch movements described in this section are associated with. While their distribution is tightly tied to words, the syllables that are targeted differ from those marked with word-level metrical prominence.

These observations suggest that Northern Pomo’s “lexical pitch-accent” conflates independent but overlapping phenomena rather than representing a unified system. First, word-level metrical structure contributes increases in duration to even syllables in monomorphemic words composed of light syllables (Dailey 2023). Second, a *default-to-opposite* pattern marking the left-most closed syllable, or else the right-most syllable, with a f0 peak. Crucially, these two systems of prominence assignment do not align: in *ʃadodo* ‘wood owl’, rhythmic durational increases target the second syllable while the f0 peak appears on the third and final syllable.²⁵ If the pitch movements were part of a “lexical pitch-accent” we might expect the location of pitch peaks to be unpredictable and lexically specified. Instead, their distribution is predictable from syllable structure, though the complexity is increased by phonemically long vowels which interrupt the general pattern with a low or falling contour. At present, it remains unclear which level of prosodic structure these pitch movements are associated with; their distribution is tied to words, but the *default-to-opposite* system targets syllables that differ from those marked with word-level metrical prominence.²⁶

Additionally, Halpern (1984, fn. 34) and Shepardson (2008) point to a small number of words and morphemes which, despite having phonemically long vowels are realized with high or rising f0 contours and suggest that Northern Pomo may exhibit lexical tone. The presence of lexical tone contrasts in Northern, however marginal, is further motivated in the following section (3.4) through the presentation of (near-)minimal pairs which contrast in terms of High vs. Low f0.

²⁵ It is also worth noting that realization of the f0 peak on the third syllable of *ʃadodo* ‘wood owl’ could be due to “peak delay” a cross-linguistically well attested phenomenon in which the f0 peak of a rising accent is realized after the syllable it is meant to target (see Arvaniti, Ladd & Mennen 1998 for Greek, Prieto, Santen & Hirshbert 1995 for Spanish, and Xu 2001 for Mandarin Chinese). In this case, the metrically strong second syllable could well be the intended target for the pitch peak with its realization in the third syllable being due to peak delay. More work needs to be done in order to rule this possibility out.

²⁶ This observation may be related to “metrical incoherence” in which languages exhibit evidence of incongruent prominence systems at different prosodic levels (Gordon 2016). Metrical incoherence has been claimed to be a feature in both Southern Pomo (Kaplan 2022) and Kashaya (Buckley 2023).

3.4 Evidence from tonal minimal pairs

3.4.1 Tonal contrasts on CVV syllables

The data presented in Section 3.3 above demonstrate the strong relationship between syllable structure and f₀ in Northern Pomo. However, this is not wholly predictable. Figure 5 shows the waveform and pitch track for two monosyllabic CVV words - *má:* ‘you (pl.)’ vs. *mà:* ‘ground, earth; things’.²⁷ In each case the vowel is segmented with two black lines indicating its boundaries; both words were produced by Edna Campbell Guerrero and the pitch tracks are plotted in a range from 100 Hz to 175 Hz so that f₀ contours may be visually compared across utterances.

In Figure 5, both *má:* ‘you (pl.)’ (left) and *mà:* ‘ground, earth; things’ are composed of the same segments, a bilabial nasal /m/ and a low central vowel /a/. The total duration of each word is about 440 ms. The vowel in *má:* ‘you (pl.)’ is approximately 346 ms in duration while the vowel in *mà:* ‘ground, earth; things’ is approximately 322 ms, showing that they are both phonemically long. However, in *má:* ‘you (pl.)’, there is an average f₀ of 146 Hz while in *mà:* ‘ground, earth; things’ there is an average f₀ of 121 Hz. making differences in f₀ the primary phonetic difference between these two words.

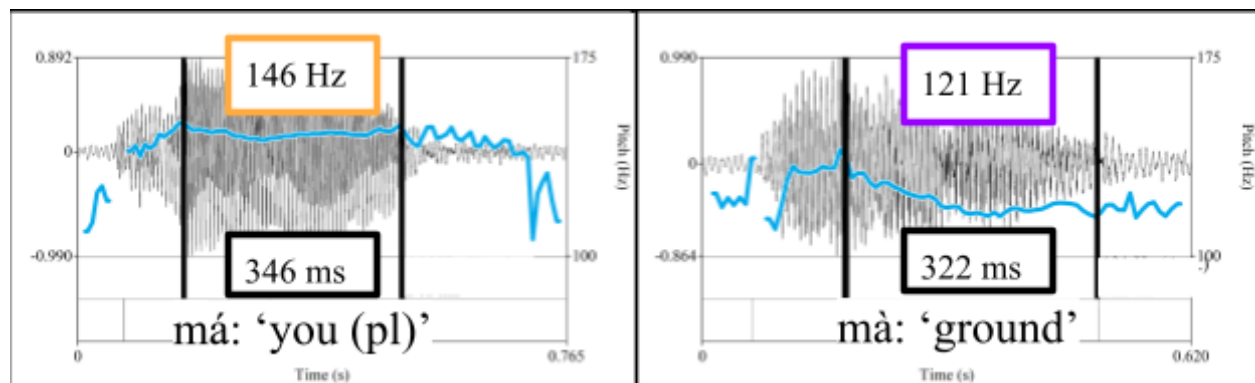


Figure 5: Pitch track and waveform for prototypical examples of *má:* ‘you (pl.)’ (left) vs. *mà:* ‘ground, earth; things’ (right) in isolation as spoken by Edna Campbell Guerrero.

Although tone is contrastive on heavy syllables in Northern Pomo, there is a strong correlation between phonemically long vowels and L tone (Vihman 1976; O’Connor 1992, pp.

²⁷ The representative examples of *má:* ‘you (pl.)’ and *mà:* ‘ground, earth; things’ shown in Figure 5 were produced by Edna Campbell Guerrero. The sound file was found on the Talking Dictionary page on northernpomolanguagetools.com. I have not tracked down the exact elicitation that it is from. However, it was likely uttered in isolation during a word-list elicitation.

8-9; Shepardson 2008). In the *Proto-Pomo* word-list, McLendon (1973) cites 72 Northern Pomo forms with phonemically long vowels that are attested in the recordings collected by Vihman and O'Connor: 68 (94.4%) are attested with L tone and 4 (6.6%) are attested with H tone. In fact, only seven potential words with H bearing phonemically long vowels have been identified in Northern Pomo so far. These are displayed in Table 3 with an accompanying L toned minimal pair:²⁸

Long vowels with H tone	Long vowels with L tone
<i>bá:</i> 'man'	<i>(ha)bà:</i> 'tongue'
<i>bitá:</i> 'bear'	<i>baṭà:</i> 'shoulder'
<i>bó:</i> 'here'	<i>bò:</i> 'west'
<i>dilé:</i> 'all'	<i>dilè:</i> 'between'
<i>má:</i> 'you (pl.)'	<i>mà:</i> 'ground, things'
<i>mú:</i> 'there'	<i>bù:</i> 'bead necklace'
<i>jó:</i> 'under'	<i>jò:</i> 'south'

Table 3: (Near-)minimal pairs in Northern Pomo demonstrating the contrast between H and L tone on CVV syllables.

3.4.2 Tonal contrasts on CVC syllables

Figure 6 shows the waveforms and pitch tracks for two phrases: *mùl tʰót' kat'a* 'that tastes rotten' and *mùl mòts' kat'a* 'that tastes sour'.²⁹ These phrases place two CVC words with ejective coda consonants - *tʰót'* 'rotten' and *mòts'* 'sour' - in the same phrasal context, allowing for their comparison. The vowels under comparison are segmented with thick black lines; both

²⁸ *bá:* 'man', *bitá:* 'bear', *má:* 'you (pl.)', and *mú:* 'there' are uncontroversial examples of words with long vowels and high f0; *bó:* 'here', *jó:* 'under', and *dilé:* 'all' have been previously analyzed as having closed syllables composed of a phonemic short vowel and a glide (*bów* 'here', *jów* 'under', and *diléj* respectively). An instrumental analysis of vowel quality is necessary to help settle the issue.

²⁹ Both phrases *mùl tʰót' kat'a* 'that tastes rotten' and *mùl mòts' kat'a* 'that tastes sour' were produced by Elenor Stevenson-Gonzales in an elicitation led by Cathy O'Connor on January 1st, 2002.

phrases were produced by Elenor Stevenson Gonzales and f0 is plotted within a range from 100 Hz to 250 Hz so that the f0 contours may be visually compared across phrases.

Comparing the target words across the two utterances, the average f0 in *tʰót* ‘rotten’ (214 Hz) is much higher than the average f0 in *mòts* ‘sour’ (156 Hz). Additionally, both target words happen to be in the same phrasal context - following *mùl* ‘that’ which is specified for L tone and preceding *kat’a* ‘to taste’ which is not specified for tone. In *mùl tʰót kat’a* ‘that tastes rotten’, there is a large step up in average f0 from *mùl* (167 Hz) to *tʰót* (214 Hz), with the pitch track showing a high-level f0 contour on *tʰót* ‘rotten’; the sharp fall at the end of the vowel being an artifact of the following ejective consonant. In *mùl mòts kat’a* ‘that tastes sour’ there is a fall in f0 from *mùl* (167 Hz) to *mòts* (156 Hz), with the pitch track showing a low-level f0 contour on *mòts* ‘sour’; again, the sharp fall at the end may be due to the following ejective. In both utterances, the pitch track shows a low f0 on *mùl* ‘that’ which rises through the onset and falls through the coda. In both utterances, the f0 on the final word *k’ata* ‘tastes’ shows the rising contour characteristic of words composed of CV syllables outlined in Section 3.3.

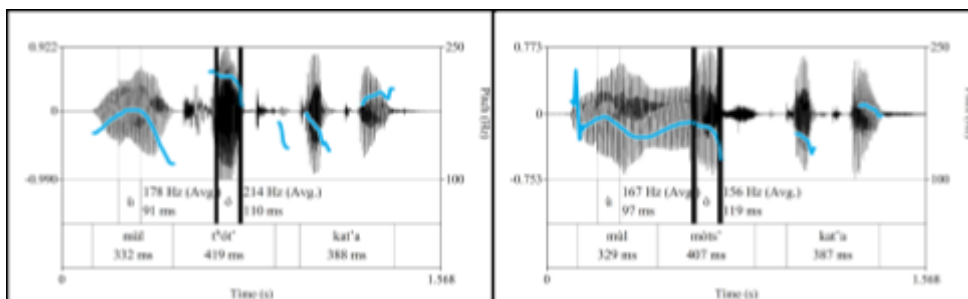


Figure 6: Pitch track and waveforms for *tʰót* ‘rotten’ (left) and *mòts* ‘sour’ (right) spoken by Elenor Stevenson Gonzales in the phrase *mùl X kat’a* ‘that tastes X’.

3.4.2.1 L tone on CVC syllables and closed syllable shortening

Northern Pomo disallows phonemically long vowels in closed syllables. In cases where a morphological context would derive a CVVC syllable, the long vowel is realized phonetically as a short vowel, a process called *closed syllable shortening*. However, closed syllable shortening does not affect the syllable’s tonal association. In all cases of which I am aware, when a CVC syllable that is specified for L tone is pronounced in a context where the syllable is open, the vowel surfaces as a long vowel. This is demonstrated with the relationship between examples

mò:ts ‘sour’ in (1a) and *mò:ts’amanhan* ‘not becoming sour’ in (1b).³⁰ These forms are presented in a four line morphogloss. The first line represents a broad phonetic transcription with periods indicating syllable boundaries. The second line represents a morphophonemic transcription showing the underlying form of each morpheme. The third line presents a gloss and the fourth line presents a free translation:

- | | |
|------------------|-----------------------|
| (1) a. [mò:ts’.] | b. [mò:ts’a.man.han.] |
| /mò:ts’/ | /mò:ts’-ama-nha-n/ |
| sour | sour-CONT-NEG-PROG |
| ‘sour’ | ‘not becoming sour’ |

In (1a), /mò:ts’/ ‘sour’ is not suffixed with an overt morpheme and surfaces as a closed syllable with L tone [mò:ts’.] as is shown in Figure 6. However, in (1b), /mò:ts’/ ‘sour’ is inflected as a derived verb with the vowel initial ‘continuous’ suffix *-ama*.³¹ As such, the final consonant is parsed as the onset of the following syllable and the L toned vowel surfaces in an open syllable, allowing for it to be realized as a long vowel: [mò:ts’a.man.han.]. While this opens up the possibility that all vowels in CVC syllables with L tone are underlying long, there are a number of words in which the concerned vowel may never surface in an open syllable. This is demonstrated with the adverbial *tʃalil* ‘just’ in (2a) and the case marking paradigm for the noun *tʃũ?* ‘person’ in examples (3a-b):

- (2) a. *tʃalil* ‘just’

- | | | |
|-------------------|-------------------|-------------------|
| (3) a. [tʃũ?.jaʔ] | b. [tʃũ?.ja.tʃũl] | c. [tʃũ?.ja.tʃũʔ] |
| /tʃũ?.jaʔ/ | /tʃũ?.ja.tʃũl/ | /tʃũ?.ja.tʃũʔ/ |
| person-NOM | person-ACC | person-OBL |
| ‘Person (nom.)’ | ‘person (acc.)’ | ‘person (obl.)’ |

The adverb *tʃalil* ‘just’ in (2a), is never observed with a suffix in Northern Pomo. Additionally, the paradigm in (3a-b) shows *tʃũ?* ‘person’ suffixed with the ‘nominative’ *-jaʔ*, ‘accusative’ *-jatʃũl*, and ‘oblique’ *-jatʃũʔ* suffixes. The word *tʃũ?* ‘person’ may also be

³⁰ The form *mò:ts’amanhan* is attested in the narrative “Preparation of Sourdough Acorn Bread” told by Annie Lake and recorded by Eero Vihman in the 1960s. I do not know the exact date. This narrative is available under password protection on the Stories and Texts page of *northernpomolanguagetools.com*. The full phrase that it is attested in is (*nan meʔ*) *meʔ min ʔà:mi-kaʔmeʔ dishiy kay meʔ min naʔ nakan min ʔà: meʔ naʔkaʔe meʔ k'o mò:ts’amanhan tʰot’in*. The free translation provided is ‘That kind of thing my grandmother’s people also made, so I thus tried but it **did not sour**, rotting’. The free translation in (1b) is adapted from this.

³¹ The typical form of this suffix is *-ma* with an initial consonant. It is not immediately apparent what the source of the preceding *a* is.

immediately followed by the determiners =*nàm* ‘the’ and =*náj* ‘that’. Notably, each option is consonant initial. As such, the CVC syllables with L tone in *tʃalɪl* ‘just’ and *tʃɛ?* ‘person’ will never surface in a context where the coda may resyllabify and phonemic vowel length is rendered opaque. As the tonal minimal pairs in Table 3 show that tone is not predictable on long vowels anyway, the motivation for analyzing these as underlyingly CVVC syllables is diminished.

3.4.3 Long vowels without specified tone

Finally, the word *ma:ju* ‘dove’ has a phonemically long vowel that may not have a tonal specification. This is shown in Figure 7 with a comparison of the f0 contours on *ma:ju* and the similarly structured word *mà:ʒa* ‘woman’.^{32, 33} Both words are presented followed by two additional L toned morphemes, the determiner clitic *nàm* and a case marking clitic *mò:w* ‘3msg.NOM or *mà:n* 3fsg.NOM. A four line morphogloss for each word is presented in example (1). *ma:ju* ‘dove’ is produced by Annie Lake while *mà:ʒa* ‘woman’ is produced by Edna Campbell Guerrero so the f0 values in the pitch tracks may not be directly compared. However, the relevant point is the difference in the f0 contour moving from the of the first syllable outlined in black to the second syllable outlined in red.

The left hand panel of Figure 7 shows the waveform and pitch track for *ma:junàmò:w* ‘the dove’. The vowel in the first syllable is about 183 ms in duration with an average f0 of 193 Hz. The vowel in the second syllable is about 80 ms in duration and an average f0 of 207 Hz. The pitch track shows a relatively flat f0 over the duration of the first syllable with a rise up to a peak in the second syllable before falling to 184 Hz in the final syllable. The right panel of Figure 11 shows *mà:ʒanàmà:n* ‘the woman’. The vowel in the first syllable is about 185 ms in duration with an average f0 of 157 Hz. The vowel in the second syllable is about 45 ms in duration with an average f0 of 154 Hz. The pitch track shows a relatively flat f0 contour from the first syllable of *mà:ʒanàmà:n* ‘the woman’ through the second syllable, gradually falling to an average of 149 Hz on the final syllable.

³² Attestation of the form *ma:junàmò:w* ‘Dove’ comes from the phrase *ma:junàmò:w kadema sili: ba-kà? k'alà:n* ‘The dove, after going to play grass game, his grandmother died’ in the narrative ‘Why Dove Cries’ as told by Annie Lake.

³³ Attestation of *mà:ʒanàmà:n* comes from the phrase *mà:ʒanàmà:n kawinàm sip'ù:ni* ‘the woman kissed the child’ elicited by Cathy O'Connor with Edna Campbell Guerrero on December 1st 1979.

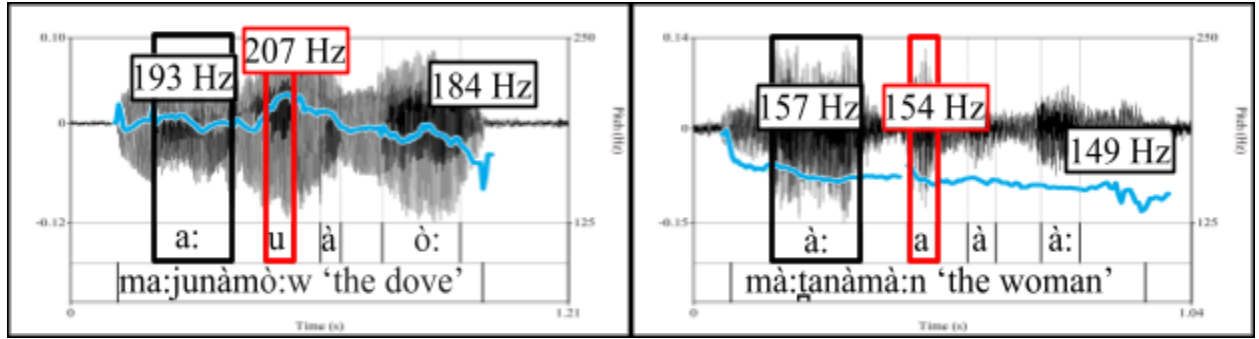


Figure 7 *ma:junàmò:w* ‘Dove’ (top) as produced by Annie Lake and *mà:tanàmà:n* ‘the woman’ as produced by Edna Campbell Guerrero.

As with *fakò:do* ‘rabbit’, f_0 is low on the CVV initial syllable in *mà:ta* ‘woman’ and remains low through the second syllable as it transitions into the clitics $=nàm=mà:n$ ‘the=3fsg.NOM’. However, even though *ma:ju* ‘dove’ has a CVV initial syllable, f_0 steps up to its peak on the second syllable before falling through $=nàm=mò:w$ ‘the=3msg.NOM’. The f_0 contour in *ma:ju* ‘dove’ mirrors the f_0 contour observed on words composed of CV syllables (see Section 3.3) which also exhibit their characteristic rising f_0 when followed by morphemes with L tone such as the determiner enclitic $=nàm$ ‘the’. If the first syllable were specified for a L tone, as I propose for *mà:ta* ‘woman’, the f_0 contour over the second syllable would be expected to remain low-level or slightly falling through the second syllable into the following clitics. On the other hand, if the initial syllable was specified for a H tone, the f_0 contour might be expected to hit its peak in the first syllable with a sharp fall into the following clitics.

3.4.4 Level tones or early vs. late alignment?

Given the relationship between f_0 and phonemic vowel length discussed in Section 3.3, as well as earlier implications of “lexical pitch-accent” (McLendon 1973; Vihman 1976), it might be tempting to analyze Northern Pomo’s tonal contrasts in terms of an early vs. late alignment of a H or HL accent as has been proposed for certain dialects of modern Swedish (Bruce 1977, and discussion in Section 2.4). In the relevant dialects of Swedish, Accent 1 is represented with an early aligned HL tonal sequence, with the L coinciding with the stressed syllable (HL*) and Accent 2 is represented with a late aligned HL tonal sequence, with the H coinciding with the stressed syllables (H*L). When uttered in isolation, the H tone of Accent 1 is truncated so that the word is realized with low pitch on the first syllable. However, when realized

in context, the H tone is realized with a pitch peak at the end of the word preceding the stressed syllable, and a fall into the L on the stressed syllable.

To represent Northern Pomo's tonal contrast in a similar way, the CVC and CVV words with apparent L tone (e.g. *mòts* 'sour' and *mà:* 'ground') would be specified with an early aligned HL* sequence and the realization with a low-level f0 contour would be due to truncation of the H tone in isolation. On the other hand, the CVC and CVV words with apparent H tone (e.g. *tʰót* 'rotten' and *má:* 'you (pl.)'), would be specified with a late aligned H*L, leading to the high or rising f0 contour. However, I favor an analysis with level-tones for a couple of reasons.

First, the right hand panel of Figure 6 shows L toned words *mùl* 'that' and *mòts* 'sour' in succession. If the tonal contrast were represented through the alignment of a HL tonal sequence, there should be some phonetic indication of the leading H tone, such as an f0 peak, intervening between the two words. However, the pitch track shows a falling f0 contour from *mùl* to *mòts* with no intervening peak. As such, it seems appropriate to analyze the low or falling f0 contour associated with *mà:* 'ground, earth; things' and *mòts* 'sour' (and even the low or falling pitch on CVV syllables in polysyllabic words displayed in Section 3.3) as being the realization of a level L tone. Additionally, and as discussed in Section 3.3, the high or rising f0 contour observed on words composed of CV syllables appears to target the word edge while the low or falling f0 associated with many phonemically long vowels appears to target the CVV syllable. If this were a contrast in the alignment of a H or HL sequence relative to the prominent syllable, *fadodo* 'wood owl' (Figure 4) might be expected to have a peak in f0 on the second syllable and fall through the final syllable (though see footnote 27). Instead, the peak is realized on the third and final syllable, not the metrically prominent second syllable.

While the existence of minimal pairs like those discussed in this section is suggestive of a H vs. L lexical tone contrast, the highly predictable nature of their distributions makes it difficult to state which tones are lexically specified and which are assigned phonologically. Either way, the result is a surface contrast between level high and level low f0 on certain CVV and CVC syllables.

3.5 Tone and accent in the Pomoan languages

Thus far, none of the Pomoan languages have been described with lexical tone. Instead, these languages have been described as having a word-level prosodic feature that is referred to as

the “accent” and is realized either through a constellation of features referred to as “stress”, a raised pitch, or both “stress” and a raised pitch (McLendon 1973:33-35; Halpern 1984:29). Although the prosodic systems have not been described in detail, what description is available indicates that “stress” here tends to indicate an increase in relative loudness. For example, McLendon (1975) states that in Eastern Pomo, “primary stress consists of relative loudness and a shift in pitch” (p. 13). Similarly in Southern Pomo, Halpern (1984:34) discusses “stress” in terms of loudness.³⁴ In Central Pomo, Halpern (1984) describes the “accent” as being “primarily stress, with some accompanying rise in pitch” (p. 29) while Mithun (2018) states that “lexical stress... is realized as higher pitch” (p. 122). In Kashaya, the “accent” is realized as a “raised pitch”. Although I have argued that the pitch movements seen in Northern Pomo represent the confluence of lexical and boundary tones, the distinctive pitch movements have traditionally been viewed as the exponents of “accent” in Northern Pomo (Vihman 1976; Halpern 1984:29; Shepardson 2008).

In connection with Northern Pomo’s L tone, Kashaya’s “raised pitch” accent is reported to avoid phonemically long vowels (Oswalt 1961; 1988) in a process called “Accent Shift” (Buckley 2019, or “Stress Shift” Buckley 1994ab). Although Kashaya’s Accent Shift has not been discussed in terms of pitch or tone, the parallels between this process and Northern Pomo’s L tone are striking. In particular, Northern Pomo words with L bearing CVC syllables directly correspond to Kashaya words that trigger Accent Shift. This is shown in Table 4 below:³⁵

³⁴ Walker (2020:49) suggests that pitch is the primary correlate of stress in Southern Pomo. However, Halpern’s (1984) description that “in [Southern Pomo]... loudest stress accompanied by raised pitch, both with falling contour, occurs on the penult of a breath-group” (p. 38, emphasis mine), leaves open the possibility that pitch is a feature of phrasal prominence and not word-level prominence.

³⁵ With the exception of *mo:s* ‘sour’, the Kashaya forms cited here are adapted from Buckley (2019). I present these forms with underlying vowel length following the convention of Oswalt (1961) and Buckley (1994b). Buckley (2019) argues for the phonemic status of Accent Shift instead of underlying vowel length. I also use *tʃ* and *tʃʰ* where Buckley and Oswalt use *c* and *cʰ*.

Northern Pomo	Kashaya
<i>mòts</i> ‘sour’	<i>mo:s</i> ‘sour’
<i>tʃâ?</i> ‘person’	<i>ʔatʃa:tʃ</i> ‘person’
<i>hètʃ</i> ‘claw, nail’	<i>he:tʃ</i> ‘claw, nail’
<i>tʃalil</i> ‘just’	<i>tʃale:l</i> ‘random’

Table 4: Correspondences between Northern Pomo CVC syllables with L tone and Kashaya words that trigger Accent Shift

Additionally, Kashaya exhibits the same process of closed syllable shortening discussed for Northern Pomo in Section 3.4.2.1. As a consequence, Accent Shift is the only clue to the underlying vowel length in Kashaya’s CVVC syllables.³⁶ The parallels in the distribution of Northern Pomo’s L tone and Kashaya’s Accent Shift may suggest a common diachronic source, even if Accent Shift is not realized in terms of low pitch.

The reconstruction of the “accent” to Proto-Pomo is relatively uncontroversial (McLendon 1973; Moshinsky 1976; Halpern 1984; Buckley 2014). Its location is implicated in diachronic processes of vowel and syllable loss in all of the Pomoan languages (McLendon 1973:35-41) and the relative location of the “accent” is reconstructable to either the first or second syllable of the proto-form (McLendon 1973) with a strong preference for the second syllable (Moshinsky 1976; Buckley 2014). In order to account for the dual realizations, McLendon (1973) suggests that it could be characterized as a kind of “pitch-stress accent” (p. 34) which I will refer to as the historical Pomoan Accent and its reflex in the daughter languages as the Accent for the remainder of this paper.³⁷ In the following sections, I will argue that the pitch component of this Accent played a pivotal role in Northern Pomo’s tonogenesis.

³⁶ In fact, Buckley (2019) argues that Accent Shift should be considered phonemic in CVC syllables that are never observed with a phonemically long vowel.

³⁷ It is likely that these “pitch” and “stress” components of the accent may be teased apart into at least two separate phonological units: (1) word-level metrical structure (or lexical stress), responsible for the diachronic loss of syllables, and (2) a pitch-accent that is attracted to the stressed syllable, responsible for the pitch modifications that are reported to accompany stress in the daughter languages. However, the necessary comparisons to reconstruct this are beyond the scope of this paper.

4. The development of tone in Northern Pomo

This section outlines the correspondences of Northern Pomo's H and L tones across the Pomoan language family. As I will show, Northern Pomo's L tone consistently corresponds to phonemically long vowels in the attested forms of the Western Pomo languages. Furthermore, each long vowel which comes to have a L tone in Northern Pomo may be reconstructed to an accented position in Proto-Western Pomo. On the other hand, each phonemically long vowel with a H tone in Northern Pomo corresponds either to a disyllabic *aja* sequence, or a vowel plus glide sequence in the other Western Pomo languages.

The comparative data cited in this section is largely adapted from the list of 350 morphemes provided in McLendon's (1973) *Proto-Pomo* with minor revisions made by Gene Buckley (p.c.).³⁸ The Proto-Western Pomo reconstructions are proposed by me and are defended in the discussion of each cognate set. The data reported for Northern Pomo is transcribed to reflect the forms as they are attested in the audio corpus collected by Vihman and O'Connor. Following McLendon (1973), the location of the accent is not marked in the forms reported for Kashaya and Southern Pomo. This is ostensibly due to the fact that the location of the Accent is predictable in both languages. However, this omission may be further motivated by the fact that Kashaya's "accent" is a feature of a constituent which is larger than a word (Oswalt 1961; 1988) and that Southern Pomo's prosodic system has undergone extensive restructuring, likely through influence from contact with neighboring Coast Miwok speakers (Buckley 2014).

Table 5 shows correspondences for Northern Pomo's H and L tone across the language family using the cognate sets for the (near-)minimal pairs - Pn *mà*: 'earth, ground; things', *má*: 'you (pl.)', *mòts* 'sour' and *tʰót* 'rotten', as well as the tonally unspecified word Pn *ma:ju* 'dove'. As I will argue below, the development of tonal contrasts in Northern Pomo proceeded through three distinct sound changes. (1) The development of L tone on accented CVV syllables. (2) The contraction of **aja* sequences and the vocalization of glides in **ej* and **ow* sequences led to the development of new long vowels which had not developed L tone. And, (3) phonemically long vowels were shortened in closed syllables without changing their tone leading to new CVC syllables with L tone.

³⁸ Changes in Orthographic convention were made in an attempt to make the transcriptions align more closely with the IPA. The changes from McLendon(1973) are as follows *j* = *y*, *ʃ* = *š*, *tʃ* = *č*, *ts* = *c*, *t̚* = *t*, and *t̚* = *ṭ*. In the marking of stress ' is used instead of the acute accent symbol ´. The acute accent symbol ´ is used to indicate high pitch in Northern Pomo and the raised pitch "accent" in Kashaya. The grave accent symbol ` is used to indicate low pitch in Northern Pomo.

In this section I will follow the practice of using the full name of each language when discussing it or its speakers and using the following abbreviations when presenting data:

Proto-Pomo = PP, Proto-Western Pomo = PWP, Proto-Western Branch: Southern Group = PWSP, Southeastern Pomo = Pse, Eastern Pomo = Pe, Northeastern Pomo = Pne, Northern Pomo = Pn, Kashaya = Pk, Southern Pomo = Ps and Central Pomo = Pc. Section 4.1 deals with the development of L tone and Section 4.2 deals with the development of H tone.

The correspondences for the (near-)minimal pairs shown in Section 3.4 are presented in Table 5 on the next page:

		Western Branch								
			Western Branch: Southern Group							
TONE	Gloss	Pn	Ps	Pk	Pc	PWP	Pne	Pe	Pse	PP
L	THING	mà:	ʔam:a ³⁹	ʔama:	'ma:	*ʔa'ma:	--	'ʔa:m ⁴⁰	--	*'ʔa(:)ma:
	EARTH	mà:	ʔam:a	ʔama:	'ma:	*ʔa'ma:	'ʔama:	'mar	'mat	*ʔah'mat
	SOUR	mòts'	mo:ts'	mo:s' ⁴¹	'mo:s'	*'mo:ts'	--	'mo:ts'	--	*'mo:ts'
∅	DOVE	ma:ju	ma:ju	ma:ju	ma:'ju: ⁴²	*ma:'ju	ma'ju:	ma:'ju:	'mju	*ma:'ju(:)
H	YOU PL SUB	má:	ʔa:maja	ʔamaja	'maja	ʔa'maja	'ʔamaj	'ma:	--	ʔa'maja
	ROTTEN ⁴³	t ^h ót'	ʔiht ^h ot'	ht ^h ot	't ^h ot'	*ʔih't ^h ot	--	--	--	--

Table 5: Correspondences for Northern Pomo's H and L tones.

4.1 On the development of Low tone in Northern Pomo

Starting with the forms attested with L tone in Northern Pomo, Pn *mà:* 'earth, ground; things' represents a convergence of the cognate sets for EARTH and THING. Pn *mòts'* represents the cognate set for SOUR. In all three cases, the L bearing vowel corresponds to a phonemically

³⁹ The typical correspondence of word final vowel length in Southern Pomo is the loss of vowel length and gemination of the preceding consonant. If the preceding syllable had a laryngeal (e.g. ʔ or h) coda consonant, instead of gemination, there was metathesis of the onset and the coda corresponding to the lack of word final vowel length. In cases where McLendon (1973) reconstructs a *CVCVV syllable, Halpern (1984) reconstructs *CVCCV or *CVCHV corresponding more closely to the observed form in Southern Pomo. He suggests that the development of low pitch on Northern Pomo stems with phonemically long vowels is tied to the shift from the *CVCCV or *CVCHV shape in Proto-Pomo to (CV)CVV in Northern Pomo. I will not directly engage with Halpern's (1984) analysis here.

⁴⁰ Length developed independently in the first syllable of Eastern Pomo words inherited from Proto-Pomo. As such, the presence of a long vowel in the retained initial syllable might not have been inherited from Proto-Pomo.

⁴¹ McLendon reports Pk *mos'* without phonemic vowel length. I have opted to include phonemic vowel length in Pk *mo:s'* SOUR reported here is based on a correspondence from Buckley stating that Pk *mos'* SOUR triggers accent shift. This word is not included in the data presented by Buckley (2019); Oswalt (1961) reports Pk *mos'* without phonemic vowel length; however, in this section, Oswalt also presents a number of other CVC forms without phonemic vowel length that Buckley (2019) reports as triggering Accent Shift e.g. *he:c'* 'nail' and *k'i:s* 'red' Buckley (2019, exs. 22a & 23a)

⁴² McLendon (1973) suggests that the Pc form *ma:'ju:* is a borrowing from Pe. The expected reflex of PP **m* in Pc is *b* before *a*.

⁴³ The cognate set for ROTTEN is not reported in McLendon (1973). The forms cited here were found in Oswalt's unpublished Southern Pomo Dictionary, Kashaya Dictionary, and Central Pomo Dictionary. I have not been able to find related forms in any of the non-Western Pomo languages. However, this is due to a lack of effort on my part and should not suggest that they do not exist.

long vowel in the other Western Pomo languages: THING & EARTH Pn *mā:* - Ps *ʔam:a*, Pk *ʔama:*, Pc *'ma:*; SOUR Pn *mōts* - Ps *mō:ts*'; Pk *mō:s*';⁴⁴ Pc *'mō:s*'. These forms support the reconstruction of the following forms to Proto-Western Pomo: THING & EARTH PWP **ʔa'ma:* and SOUR PWP *'mō:ts*'. The reconstruction of the historical Accent on the second syllable in the Proto-Western Pomo form for THING & EARTH is supported by the loss of the ʔv- initial syllable in Northern and Central Pomo.

The reconstruction of word final *-t* in the Proto-Pomo form for EARTH PP **ʔa(h)'mat* is based on the non-Western Pomo forms Pe *'maɫ* and Pse *'mat*. There are a number of other forms for which a phonemically long vowel may be reconstructed to Proto-Western Pomo but a short vowel followed by a consonant is reconstructed to Proto-Pomo. An example for each of the other consonants is presented in Table 6 below. As shown with the three reconstructions: PP ARM **ʔi'faɫ*, and TWO **ʔa'q^hots* and EARTH PP **ʔa(h)'mat*, there is a small set of consonants, **-p*,⁴⁵ **-t* and **-ts* that are reconstructable to the level of Proto-Pomo, but develop into length of the preceding vowel in word final positions. These reconstructions are supported by the fact that consonants are retained in Eastern and Southeastern Pomo but lost in Northeastern Pomo and all of the Western Pomo languages. This indicates that vowel length at the level of Proto-Western Pomo and not Proto-Pomo is the relevant factor in the development of L tone in Northern Pomo.

	Northern Pomo	Proto-Western Pomo	Proto-Pomo
ARM	<i>fā:</i>	<i>*ʔi'fa:</i>	<i>*ʔi'faɫ</i>
TWO	<i>xò:</i>	<i>*ʔa'q^ho:</i>	<i>*ʔa'q^hots</i>

Table 6: Phonemically long vowels in Northern Pomo corresponding to short vowels followed by certain word final consonants in Proto-Pomo.

⁴⁴ On the surface, the Kashaya form *mo:s* 'rotten' would undergo closed syllable shortening and be realized with a short vowel. The form reported here includes a phonemically long vowel in part to indicate that it would induce Accent Shift without needing to introduce a new symbol and in part to make the correspondence between L tone in Northern Pomo and phonemic vowel length across the Western Branch explicit.

⁴⁵ *p* indicates a proto-consonant of unknown value that developed into vowel length in the Western Pomo languages (plus Northeastern Pomo) but *-l* in Eastern and Southeastern Pomo in word final position. This is contrasted with **-l* which develops into *-n* at the end of words in Southern Pomo, and *-l* in word final position in all other Pomoan languages.

4.1.1 Low tone and accent

In Section 3.4.4, I argue that the Northern Pomo reflex for DOVE: Pn *ma:ju* has a phonemically long vowel without a tonal specification.⁴⁶ Looking across the cognates in Table 5, the vowel in the first syllable of Pn *ma:ju* corresponds to a phonemically long vowel within the Western Pomo languages: Pk *ma:ju* and Ps *ma:ju*, with the Central Pomo form Pc *ma: 'ju:* being a suspected loan from Eastern Pomo (see footnote 44). The Kashaya and Southern Pomo forms offer little insight into the location of accent at the Proto-Western Pomo level. However, the reconstruction of the historical Pomoan Accent to the second syllable is supported by the non-Western Pomo forms Pne *ma 'ju:* ; Pe *ma: 'ju:* and Pse *'mju:* where the Southeastern Pomo form reflects the loss of the first syllable's vowel in a pre-tonic syllable. The form is reconstructed to Proto-Pomo as DOVE PP **ma: 'ju(:)* and barring evidence to the contrary, I assume that this was inherited into Proto-Western Pomo with DOVE PWP **ma: 'ju* with a phonemically long vowel in the first syllable and the Accent on the second syllable.

As such, the development of L tone on Pn *mà:* 'ground, earth; things' (< PWP **ʔa 'ma:* and *mòts* 'sour'(< PWP **'mo:ts*') but the lack of tone on Pn *ma:ju* 'dove' (PWP < **ma: 'ju*) may be tied to accented long vowels at the Proto-Western Pomo level. I will call this sound change LOW TONE TO ACCENTED LONG VOWELS:⁴⁷

LOW TONE TO ACCENTED LONG VOWELS: PWP **v:* > Pn *v̀:* / ' _

A long vowel becomes a long vowel with low tone when in an accented position.

The effect of this sound change is to introduce L tone to every phonemically long vowel that is reconstructable to an accented position at the level of Proto-Western Pomo.

⁴⁶ The correspondence set for DOVE discussed in this section presents a number of issues. First, it shows irregular correspondences across the Pomoan languages, suggesting family internal borrowing. Second, it is attested with a similar form in a number of non-related languages, suggesting widespread borrowing of the form. And perhaps most problematic for the current hypothesis, Edna Campbell Guerrero is recorded mimicking the call of the mourning dove as *ma:hú:hú:* with high pitch and exaggerated length on the second and third syllables, suggesting the possibility that the form *ma:jú* 'dove' is onomatopoeic. However, the only other form that is reconstructed with a phonemic long vowel in an unstressed syllable OCCIPUT PP **k^ha: 'ja* does not have an attested reflex in Northern Pomo. Since Pn *ma:jú* 'dove' is the only attested word in Northern Pomo that has been reconstructed to a form that shows a mismatch between the reconstructed stress and the placement of the reconstructed long vowel, I will use it here for expositional purposes.

⁴⁷ The name of the sound change LOW TONE TO ACCENTED LONG VOWELS is intended to mirror the naming convention for constraints like TONE-TO-STRESS (de Lacy 2002) which motivate the association of tone to stressed syllables over unstressed syllables in OT.

While no other Pomoan languages are reported to have lexical tone, the process of Accent Shift (Buckley 2019) discussed in Section 3.5 shares striking parallels in distribution to Northern Pomo's L tone, being triggered by phonemically long vowels. If this process shares a diachronic source with Northern Pomo's L tone, it would suggest that LOW TONE TO ACCENTED LONG VOWELS operated in the Proto-Western Pomo period, before Northern Pomo and Kashaya diverged from their shared ancestor.

4.2 The development of High tone in Northern Pomo

The Northern Pomo forms *má:* 'you (pl.)' and *t^hót* 'rotten', show that Pn H tone has two different correspondences in the other Western Pomo languages.

4.2.1 High tone on phonemically long vowels

The H bearing phonemically long vowel in Pn *má:* 'you (pl.)' corresponds to a disyllabic sequence of *-aja* in across the Western Pomo languages: Ps *ʔa:maja*, Pk *ʔamaja*, and Pc *'maja*, suggesting the reconstruction of the Proto-Western Pomo form: PWP **ʔa'maja*. The reconstruction of the Proto-Western Pomo Accent to the second syllable is supported by the loss of the *ʔv-* initial syllable in Central Pomo indicating accented second syllable. At the Proto-Pomo level, McLendon, (1973, 1976) reconstructs the morphologically complex form PP YOU SUBJ - PL **ʔa'ma-ja*, composed of the independent base YOU SUBJ **ʔa'ma* and the reconstructed plural suffix PL PP **-aja*. This, in combination with the retention of the *-aja* sequence in the attested forms within the Western Branch's Southern Group suggest the reconstruction of this sequence to Proto-Western Pomo: PWP **ʔa'maja*.⁴⁸ This sound change can also be seen in the development of Pn *bá:* 'man' (< MAN (2) PWP **hiʔ'baja* < PP **hiʔbaja* 'man'). Following McLendon (1976), I will call this sound change - 'AJA CONTRACTION':

- 'AJA CONTRACTION: PWP *'aja > Pn á:

*In a disyllabic sequence of *-aja with the first vowel bearing stress, the syllables are contracted resulting in a single syllable with a long vowel a: bearing a high tone.*

⁴⁸ As the attested forms for YOU PL SUBJ in the Western Pomo languages no longer exhibit the productive pluralization strategies in each respective language, it may be the case that this form had already lexicalized in Proto-Western Pomo.

Since the formation of a phonemically long vowel in this context appears to be a Northern Pomo innovation, the H bearing phonemically long vowel in Pn *má:* ‘you (pl.)’ must have formed after the operation of LOW TONE TO ACCENTED LONG VOWELS.

As a consequence of this sound change, new phonemically long vowels, which did not develop L tone, are introduced into Northern Pomo, disrupting the predictable relationship between phonemically long vowels, stress and tone that was established by LOW TONE TO ACCENTED LONG VOWELS.

A similar sound change may be implicated in the development of BEAR Pn *bitá:* < PWP **bu 'taqa* < PP **bu 'taqa(ʔ)*. However, more comparative work is necessary to determine if this was a regular sound change. Finally, a separate process of glide vocalization may have been implicated in the development of Pn *bó:* ‘here’, *dilé:* ‘all’, *mú:* ‘there’, and *jó:* ‘under’. This may be partially supported by the Kashaya and Southern Pomo correspondences for Pn *jó:* ‘under’ reported by McLendon (1973) - BELOW Pk *ʔijow* ; Ps *ʔijow*.

4.2.2 High tone on CVC syllables

The H bearing CVC syllable in Pn *tʰót* ‘rotten’ does not involve changes in vowel length. Instead, the corresponding forms in the other Western Pomo languages show a similar phonemically short vowel in a closed syllable: Ps *ʔihtʰot*, Pk *htʰot*, Pc *tʰot*; suggesting a Proto-Western Pomo reconstruction without a phonemically long vowel: PWP **ʔih tʰot*. Having a phonemically short vowel at the Proto-Western Pomo level means that ROTTEN PWP **ʔih tʰot* is not expected to develop a L tone while SOUR PWP **mō:ts* would. However, at this point, the difference in tone would still be redundant with vowel length. This cannot be considered a true tonal minimal pair until contrasts until phonemic vowel length was neutralized in closed syllables.

4.2.2.1 Tonal contrasts on CVC syllables and closed syllable shortening

In Proto-Western Pomo, there is a difference in the length of the vowels reconstructed to the closed syllables in ROTTEN PWP **ʔih tʰot* and SOUR PWP **mō:ts*. The sound change LOW TONE TO ACCENTED LONG VOWELS introduced a L tone on the stressed long vowel in SOUR PWP **mō:ts* > **mō:ts* leading to a redundant contrast between the L bearing long vowel in SOUR and the non-L bearing short vowel in ROTTEN PWP **ʔih tʰot* > **ʔih tʰot*. This contrast becomes fully tonal in Northern Pomo once a process of CLOSED SYLLABLE SHORTENING

neutralizes phonemic vowel length in closed syllables without altering the tonal specification e.g. SOUR **'mò:ts'* > Pn *mòts'*.

CLOSED SYLLABLE SHORTENING: **v:* > *v* / C_ C]_{syll}

Phonemically long vowels are shortened in closed syllables.

The fact that CLOSED SYLLABLE SHORTENING is still observed as a synchronic rule in Northern Pomo (Section 3.4.2.1) suggests that this was a recent (potentially ongoing) sound change in the history of Northern Pomo.

Of the forms displayed in Table 4, this sound change may also be demonstrated in the development Pn *tʃàʔ* 'person' (< PWP **ʔa'tʃa:tʃ* < PP **ʔa'ka:kʔ* MAN (1)) and *hètʃ* 'claw; nail' (< PWP **he:tʃ* < PP **he:ts'* CLAW; FINGERNAIL).

4.3 An argument against segmental neutralization in Northern Pomo's tonogenesis

The correspondences outlined throughout this section indicate that the development of tone in Northern Pomo is strongly tied to phonemic vowel length and the location of the reconstructed Accent at the level of Proto-Western Pomo. A L tone was assigned to every phonemically long vowel that was inherited from Proto-Western Pomo through the sound change LOW TONE TO ACCENTED LONG VOWELS. Long vowels with H tone developed later through -'AJA CONTRACTION (McLendon 1976) and the opposition between H vs. L on CVC syllables developed as CLOSED SYLLABLE SHORTENING neutralized phonemic vowel length in closed syllables.

As discussed in Section 2, segmental contrast neutralization is the most common pathway for the development of level tone. However, the development of tone in Northern Pomo is not tied to the loss of glottal consonants in coda positions, voicing contrasts in onset position, or mergers in vowel quality. Both H and L tones develop on syllables with sonorant onsets as is observed in the minimal pair Pn *mà:* 'ground, things' vs. *má:* 'you (pl.)' and the other (near-)minimal pairs shown in Table 3. Furthermore, the development of L tone is seen in syllables with voiceless onsets as evidenced by ARM Pn *ʃà:* < PWP **ʔi'ʃa:* < PP **ʔi'ʃaʔ'*. The loss of these word final consonants is not the source of tone in Northern Pomo is seen in the development of tone in words like Pn *mòts'* < PWP **'mo:ts'* < PP **'mo:ts'* where the word final

consonant is retained but the long vowel develops L tone nonetheless, as well as the host of phonemically long vowels that developed L tone and are reconstructable to Proto-Pomo.

The development of tone is not related to vowel quality as both tones are observed on the phonemically long vowels *a:*, *o:*, *e:*, and *u:* without obvious mergers in quality. And, finally, it is unlikely that tone developed through inherent differences tied to phonemic vowel length as occurred in Hu (Svantesson 1989, 1991). As discussed in detail in Section 2.3, the phonetic relationship between vowel length and f0 is tenuous at best (e.g. Connell 2002) and any effect that there may be appears to be neutralized by listeners (Yu 2010; Yu, Lee & Lee 2014). Additionally, if lower f0 in phonemically long vowels were the driving force for the development of L tone in Northern Pomo, the phonemically long vowel in *ma:ju* ‘dove’ should have also developed a L tone, but it did not. Furthermore, there is no indication that the phonemic vowel length contrast was neutralized as differences in f0 shifted to become the primary cue to the contrast. Instead, there is clear evidence that the L tone developed on accented phonemically long vowels with vowel length remaining stable until alterations to word and syllable structure created new long vowels and shortened some old ones.⁴⁹ The next section explores the phonetic and perceptual mechanisms that may have allowed the pitch modification associated with the historical Pomoan Accent to be reinterpreted as a L tone on phonemically long vowels.

5. How a pitch-accent becomes level tone

5.1 Phonetic and perceptual mechanisms underlying the pitch-accent reanalysis

As discussed in Section 2, tonogenesis typically occurs through the process of segmental contrast neutralization; automatic phonetic alterations in pitch tied to voicing or laryngeal contrasts in either the syllable’s onset or coda consonants become phonologized, replacing the contrasts which originally conditioned them (Ohala 1981, 1989, 1993; Gao & Kirby 2024). The plausibility of this proposed pathway is corroborated by the established relationship between segmental contrasts and small yet perceivable differences in f0 in nontonal languages such as American English and Arabic (Hombert, et al. 1979; Abramson 2004). However, there is no evidence of segmental contrast neutralization in the Northern Pomo’s tonogenesis.

⁴⁹ It is worth noting that the development of tonal minimal pairs in Northern Pomo is correlated with a shift towards monosyllabicity and a simplification of syllable structure which is also a key factor in the development of tone across Southeast Asia (Matisoff 1973) and Africa (Ratliff 2015 Section 16.4.2, citing Clements & Rialland 2008).

Instead of segmental contrast neutralization, Northern Pomo’s tonal oppositions develop first through the sound change LOW TONE TO ACCENTED LONG VOWELS which establishes the relationship between L tones and phonemically long vowels observed in Northern Pomo. As is briefly discussed in Section 3.5 above and again in Section 5.2.1 below, McLendon (1973) suggests reconstructing the Proto-Pomo Accent as a kind of “pitch-stress accent” (p. 34). In order to understand how a H vs. L tonal contrast can develop through an interaction between the historical Pomoan Accent and phonemic vowel length, I propose that the sound change LOW TONE TO ACCENTED LONG VOWELS be conceptualized as establishing allophony in the pitch component of this Accent, being realized as a H tone on accented short vowels and a L tone on accented long vowels. True tonal contrasts, as such, didn’t develop until changes to syllable structure through - *’AJA CONTRACTION* and *CLOSED SYLLABLE SHORTENING* rendered the relationship between the phonemic vowel length and the realization of the Accent opaque. This opacity, in turn, facilitates the reanalysis of the accentual tone as being part of the underlying representation of some words instead of being the predictable realization of the Accent.⁵⁰

The role that changes in word structure play in Northern Pomo’s tonogenesis is reminiscent of the development of tonal contrasts in the Swedish (see Section 2.4). However, the contrasts observed in modern dialects of Swedish are realized in terms of early vs. late alignment of a HL tonal melody or the number of pitch peaks realized within the word. All other cases which result in a contrast between two contrastive level tones appear to have developed through the process of segmental contrast neutralization. As such, there is no clear precedence or mechanism for an accentual tone to be phonologized as a L tone on phonemically long vowels but as a H tone on phonemically short vowels, especially when its assumed reflex in the other Pomoan languages is described in terms of a pitch peak (see Section 3.5). In the following discussion, I first present a hypothetical reconstruction of the f₀ contour associated with the historical Pomoan Accent. Then, in order to motivate a perceptually plausible account of the historical Pomoan Accent’s development into a L tone on accented long vowels and a H tone on accented short vowels, I outline current research concerning the effects of segmental material on the perception of f₀ movements (e.g. House 1990) and pitch-accent scaling (Barnes et al. 2024).

⁵⁰ Conceptualizing Northern Pomo’s tonogenesis in this way leads to the prediction that the only words which have lexical tone (i.e. tone specified in their underlying representation) are words with CVV syllables that receive H pitch (e.g. *má*: ‘you (pl.)’, *bitá*: ‘bear’) and words with CVC syllables that receive L pitch (e.g. *tʃãʔ* ‘man’, *ts’òj* ‘type of basket’). In this case, the H tones that are observed on CVC and CV syllables, as well as L tones on CVV syllables are not lexical, but assigned predictably.

5.2.1 The phonetic realization and the timing of the Proto-Pomo accent

In Kashaya, Oswalt (1961) states that the accented syllable in Kashaya has a “rise in pitch higher than its neighbors (p. 28).” Similarly, McLendon (1975) states that “pitch rises on the stressed syllable and drops on the next syllable” in Eastern Pomo (p. 13). Assuming that the realization of the “accent” in both Kashaya and Eastern Pomo is a reflection the historical Pomoan Accent, it stands to reason that the historical Pomoan Accent in the Proto-Pomo period was characterized by a rising-falling contour which reached its peak in the accented syllable and its nadir in the syllable immediately following. For this contour shape to have been inherited into Kashaya, the Proto-Western Pomo Accent would, by hypothesis, have had a similar contour as well. It is unclear whether the pitch component of the historical Pomoan Accent should be thought of as a word-level or a phrase-level phenomena; more work on the prosodic systems of the other Pomoan languages is needed before anything can be said definitively (see footnote 39). As a reminder, Swedish tonogenesis involved the lexicalization of a focus marking tone, suggesting that the kind of reanalysis proposed here is possible even in cases when the words which come to have tone are only produced with the relevant tone in certain phrasal contexts (e.g. focus).

There are wide ranging cross-linguistic reports suggesting that the initiation of accentual pitch movements and the realization of the intended targets are consistently timed with respect to segmental targets, a phenomenon known as ‘segmental anchoring’ (see Ladd 2004, 2006 for detailed overviews). For instance, Modern Greek (Arvaniti, Ladd and Mennan 1998) and Dutch (Caspers & van Heuven 1993; Ladd, Mennen & Schepman 2000) both have accents which are characterized by a low to high rising pitch contour which is initiated with the onset of the accented syllable with different alignments for the peak. The two languages, however, differ in the alignment of the H tone target. In Modern Greek, the pitch peak is aligned with the onset of the following syllable while in Dutch it is aligned with the end of the accented syllable. The realization of the pitch peak is tied to these segmental landmarks regardless of phonetic factors like speech rate or phonological factors like syllable structure.

Additionally, there is evidence that the edges of non-segmental prosodic units such as moras, syllables, feet or words may serve as anchors for the realization of f₀ targets. For instance, Prieto, D’Imperio & Gili Fivela (2005) claim that some pitch accents are phonologically specified with such associations, citing changes in categorical meaning that

depend on slight differences in alignment between otherwise identical pitch accents. To give one example, in Neapolitan Italian, a rising accent is used to signal both broad focus in declaratives and narrow focus in questions; however, the accent used to signal broad focus in declaratives is realized with the peak aligned with the end of the accented syllable whereas the accent used to signal narrow focus in questions is realized with the peak aligned early in the accented syllable. Both accents are modeled with a L+H* tonal sequence; Prieto et al. (2005) propose encoding the contrast through the different specifications for the alignment of the H* - broad focus having a H* contrastively aligned with the end of the accented syllable and narrow focus having a H* contrastively aligned with the first mora of the accented syllable. Furthermore, the alignment doesn't need to be contrastive in order to reference non-segmental units as Murphy & Yu (2007) show that H tone of a LH accent predictably anchors to the second mora of the accented syllable in Washo.

Similarly, the contour of the Proto-Western Pomo Accent - in a period before the operation of the sound change LOW TONE TO ACCENTED LONG VOWELS - could have been timed such that the peak was aligned early in the accent vowel through an association with its first mora. After reaching this target, the fall - which could be timed to reach its valley at the end of the accented syllable or somewhere in the following syllable - would pass through the remainder of the accented syllable. While the timing of the Accent's contour with respect to these landmarks would be unaffected by phonemic vowel length, crucially, the segmental material that the contour passes through would. In accented CVV(C) syllables⁵¹ the fall would pass through the end of the long accented vowel. In contrast, in accented CV(C) syllables the fall would pass outside of the syllable or through its coda consonant. This is illustrated with a stylized f0 contour drawn over CV(V)(C) skeletons in Figure 8 below.

On the left, the first vowel slot of the accented CVV(C) syllable is red indicating that the rise of the pitch contour is realized through the beginning portion of the accented vowel. The second vowel slot is blue indicating that the falling portion of the pitch contour is realized through the final portion of the accented vowel. On the right, the only vowel slot of the accented CV(C) syllable is coded red indicating the pitch rises through the entirety of the accented vowel. The falling portion of the contour is shaded grey indicating that it either passes outside of the

⁵¹ The retention of CVVC syllables in Central, Southern, Northeastern, and Eastern suggests that closed syllable shortening was not an active synchronic rule in Proto-Pomo or Proto-Western Pomo, and may be a parallel development in Northern Pomo and Kashaya.

accented syllable or through its coda, missing the accented vowel entirely.⁵²



Figure 8: Hypothetical f0 contour for the Proto-Western Pomo Accent stylized and displayed over a syllable with a phonemically long vowel (left) and a syllable with a phonemically short vowel (right).

5.2.3 Segmental effects on pitch-accent scaling

A range of studies report that listeners perceive the average height of an f0 movement as being similar to an f0 level closer to the end of the f0 movement than the beginning or the middle (Brady, House & Stevens 1961; Rossi 1971; Hombert 1976; Rossi 1978; d'Alessandro, Rosset & Rossi 1998; Barnes et al. 2024). Furthermore, Barnes et al. (2024) report that American English speakers perceive H pitch accent peaks as being relatively lower when they are realized on segments with lower sonority. Barnes et al. (2024) explain these findings through a perceptual model (Tonal Center of Gravity - Frequency) in which listeners weight f0 values relative to the timing within the pitch movement along with the sonority of the segments within which it is realized.⁵³ F0 values near the end of a pitch movement and f0 realized in more sonorant segments have a greater impact on the perception of pitch height than those at the beginning of the pitch movement or realized in less sonorant segments.

In the context of the Proto-Western Pomo Accent, the low f0 values at the end of the contour would more heavily impact the perception of pitch height when they were realized in a vowel than when they were realized in a consonant. As a result, the overall height of the Accent

⁵² A number of studies report that pitch accent targets tend to be realized relatively earlier in syllables with less sonorous rhymes (e.g. Ladd, Mennen & Schepman 2000; Prieto & Torreira 2007). Barnes et al. (2024) relate this to the cross-linguistic tendency to restrict tonal contrasts to syllables with high sonority rhymes (Gordon 2007) and suggest that these reflect a pressure for speakers to realize tonal targets in areas where the target may be more accurately perceived. This supports the idea that the pitch peak may have been realized within the vocalic portion of an accented CVC syllable in Proto-Western Pomo.

⁵³ It should be stressed that the weighting of f0 in the perception of accent height is relative to the category. A H pitch realized in a coda consonant will be perceived as less high but a L realized in the same location should be perceived as less low. Although I am assuming that the speakers in Proto-Western Pomo would have perceived the rising-falling configuration as a movement to a H target or peak and thus integrate the falling f0 into their perception of the height of the preceding peak, it is not clear *a priori* that they could not instead perceive the configuration as a movement to a L target or valley at the end of the fall and instead integrate the f0 values into the perception of the height (or relative low-ness) of the valley.

would sound lower on accented CVV(C) syllables than on CV(C) syllables. This is sketched below in Figure 9.

The color coding remains the same as in Figure 8. However, the relative heights of the stylized contours have been altered, with a dotted line drawn at their respective apex to indicate differences in the relative perceived height.⁵⁴ On the left, the blue shading of the second vowel slot and the falling portion of the contour indicate that with the fall passing through the final portion of the accented vowel, the perceptual effects of the low f0 values at the end of the contour would be exacerbated by the high sonority of the vowel that they are realized within. On the right, the grey shading of the fall indicates that the low f0 values at the end of the contour are effectively “masked”, being realized outside of the accented vowel entirely. This could be the coda consonant of the accented syllable or the onset of the syllable immediately following. If there is nothing following the accented syllable, the assumption would be that the fall is not realized. As a result, the overall height of the Accent’s f0 contour would be perceived as lower in accented CVV(C) syllables than in accented CV(C) syllables as is indicated by the blue and red dashed lines respectively.



Figure 9: Hypothetical f0 contour for the Proto-Western Pomo Accent stylized to indicate differences in perceived height and displayed over a syllable with a phonemically long vowel (left) and a syllable with a phonemically short vowel (right).

⁵⁴ As hinted at in footnote 53, I am conceptualizing the development of the allophony between a H tone accent on CV(C) syllables and a L tone accent on CVV(C) syllables as preceding through differences in the perceived height of the accent’s peak. As Barnes (p.c.) helpfully points out, if the accent is represented as a H*L with the L being realized in the second vocalic mora of an accented CVV(C), it may have been realized essentially as a falling contour. In this case, it is also conceivable that the allophony could develop through a change where the L tone is reinterpreted as the target of the accent on CVV(C). In this case, the H tone might gradually be pushed earlier into the syllable’s onset to ensure that the L tone is realized within the accented vowel. Although this is beyond the scope of the current discussion, it will be worth exploring what factors are involved in how listeners perceive the individual targets of bitonal pitch-accents, as well as the implications of an account based on a change in the perceived target as opposed to one based on changes in perceived height. It may be the case that a shift in the perceived target of the accent may be compatible and facilitated by the perceived differences in peak height discussed here.

In addition to the effect that consonants have on the perception of relative pitch height, the segmental material that a movement in f_0 passes through can affect whether it is perceived as a contour or level pitch (House 1990). The transitions between consonants and vowels are marked by disruptions to the vowel's spectral structure which make the task of processing movements in f_0 and integrating them into a percept of pitch more difficult. House (1990, 1996, 1999) presents evidence that f_0 movements passing through regions of spectral stability are more likely to be perceived as contours while f_0 movements passing through regions of spectral instability are more likely to be perceived as level pitch. In general, a long vowel, having a longer portion of spectral stability is more likely to support the perception of a pitch contour while a short vowel having a shorter window of spectral stability, is more likely to support the perception of a level pitch.

In the context of the hypothesized Proto-Western Pomo Accent, the falling portion of the accent will pass through a region of spectral stability at the end of accented long vowels, making it likely to be perceived as a falling pitch contour. On the other hand, since much of the f_0 movement in the accented short vowel will pass through periods of spectral instability, it is likely to be perceived as a level pitch. When combined with the differences in perceived pitch height discussed above, the Accent is likely to be perceived as a low-falling pitch contour when realized on a CVV(C) syllable, but as a high-level pitch when realized on a CV(C) syllable. This is schematized below in Figure 10. The stylized contours are intended to illustrate the difference between perceived pitch movements on accented CVV(C) and accented CV(C) syllables.

In accented CVV(C) syllables, the falling portion of the Accent's f_0 movement passes through the end of the accented vowel. As discussed above, this leads to the perception of an overall low pitch height when compared to accented CV(C) syllables and is indicated by the blue shading of both vowel slots in the CVV(C) syllable. In addition, the falling f_0 movement is more likely to pass through a region of spectral stability in accented long vowels, facilitating the perception of a falling pitch contour as indicated by the blue shading.

In accented CV(C) syllables, the falling f_0 passes outside of the accented vowel. Being realized either in the coda of the accented syllable or the onset of the following syllable, the low f_0 values at the end of the Accent's realization would have little effect on the perception of overall pitch height. As a consequence, pitch would sound high in accented CV(C) syllables relative to accented CVV(C) syllables. This is indicated by the red shading and the relative

height of the stylized contour over the CV(C) syllable depicted in Figure 10. In addition, the falling portion of the contour passing through regions of spectral instability would lead the Accent to be perceived as a level pitch on CV(C).



Figure 10: Hypothetical f0 contour for the Proto-Western Pomo Accent stylized to indicate differences in perceived pitch movements and displayed over a syllable with a phonemically long vowel (left) and a syllable with a phonemically short vowel (right).

As a result of the various effects of segmental material on the perception of f0 movements, the realization of the Proto-Western Pomo Accent could have been perceived as a low-falling pitch contour on accented CVV(C) syllables but as a high-level pitch contour on accented CV(C) syllables - setting the stage for reanalysis.

5.2.4 Accent reanalysis and the development of tonal contrasts

Over time, the perception of a low-falling pitch contour on accented CVV(C) syllables may have been reinterpreted as the realization of a L tone target while the perception of a high-level pitch on accented CV(C) syllables could be reinterpreted as the realization of a H target on accented short vowels.⁵⁵ In this account, the perception of falling pitch on accented long vowels is assumed to be interpreted as a redundant cue, highlighting the difference in targets on accented long vowels and accented short vowels. This is depicted in Figure 11:

⁵⁵ House's (1999) findings also show that a falling f0 movement that passes through a consonant may be perceived as a sequence of a high level pitch on the first syllable followed by a low level pitch on the second syllable. Such a sequence is consistent with the descriptions of the accent in Eastern Pomo and Kashaya.

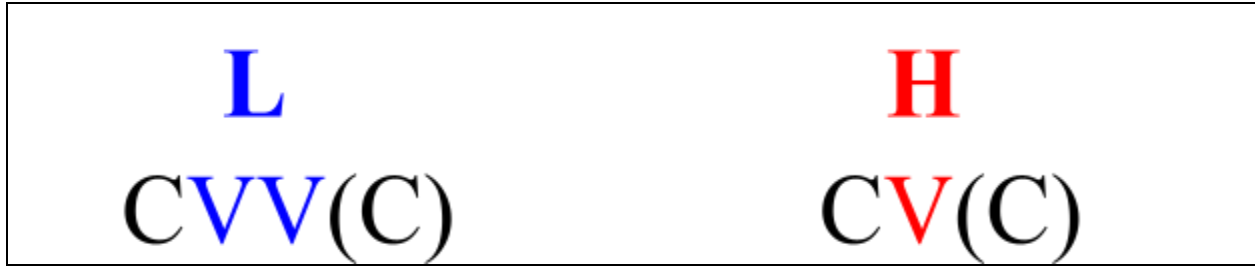


Figure 11: Reanalysis of the Proto-Western Pomo Accent into a L tone on CVVC syllable (left) and a H tone on CV(C) syllables (right).

I have discussed this change above as establishing a kind of accent allomorphy where the historical Pomoan Accent came to be realized as a L tone on CVVC(C) syllables and H tone on CV(C) syllables by the end of the Proto-Western Pomo period.⁵⁶ However, there are likely other possible alternatives. One such is that the low-falling pitch contour on CVVC(C) syllables was reinterpreted as a non-accentual L tone, maybe interpreted as an enhancement of the phonemic vowel length contrast that was associated with long vowels predictably and had the effect of blocking the regular assignment of the Accent. It is unclear if this leads to different predictions from the proposed account based accent allophony. Either way, the relationship between phonemically long vowels and L tone would have been established through this process and inherited into Northern Pomo. At this point, the distribution of tone was predictable being conditioned by vowel length: L tone on CVVC(C) syllables and H tone on CV(C) syllables. However, sound changes operating after Northern Pomo diverged from the other Western Pomo language have the effect of altering syllable structure and obscuring this relationship, resulting in tonal minimal pairs. First - 'AJA CONTRACTION (McLendon 1976) reduces historically disyllabic sequences of **aja* into a single syllable with a long vowel (schematized below on the right of Figure 12). However, during this process, the words undergoing - 'AJA CONTRACTION were still marked with a H tone accent, creating surface contrasts between L and H tone on long vowels.

⁵⁶ There is a process of vowel coalescence whereby a stem final short vowel and suffix initial short vowel merge to create a long vowel. In these cases, a L tone is associated with the derived long vowel even though the stem and suffix are typically realized with high or rising f0. For example, on its own, *p^hidi* 'to fly' has two short vowels and is realized with the expected pitch peak on the second syllable. However, when suffixed with the vowel initial durative suffix -*ade*, the stem final *i* and suffix initial *a* merge to form the long vowel *e*: which is associated with a L tone: *p^hide:de* 'flying along'. In this context, the presence of L tone doesn't appear to be part of the underlying representation of either morpheme and instead appears to be conditioned by the derived vowel length. If this is a relic of the accentual alternation after LOW TONE TO ACCENTED LONG VOWELS, it might support an analysis involving accent allomorphy over the direct development of a lexical L tone. It is also worth noting that Kashaya's Accent Shift is triggered by long vowels derived through the same process (Buckley 1994ab, 2019).

After this, CLOSED SYLLABLE SHORTENING reduced *CVVC syllables to CVC (schematized on the left in Figure 12). Again, the vowels undergoing shortening would still have been marked with L tone, leading to surface contrasts between L and H tone on CVC syllables.

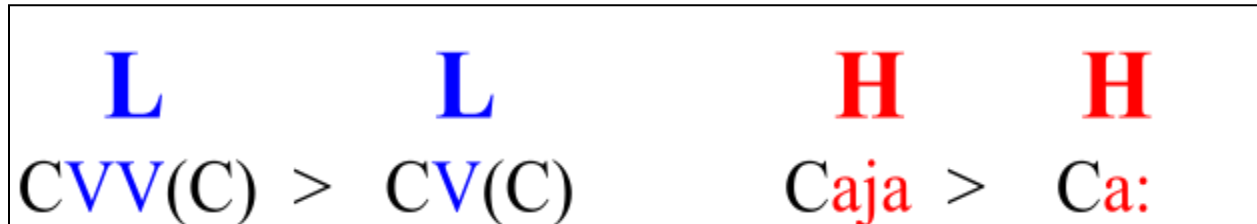


Figure12: Schematized operation of CLOSED SYLLABLE SHORTENING creating CVC syllables with L tone (left) and -AJA CONTRACTION creating Ca: syllables with H tone (right).

These sound changes have the effect of obscuring the relationship between phonemic vowel length and tone and prompting the reanalysis of the tone of the historical Pomoan Accent into the underlying representation of words in which the allophony was no longer predictable.

6. Conclusion

Northern Pomo, a dormant indigenous language of Northern California has developed level tone contrasts through the reanalysis of a pitch-accent, unlike the commonly attested and well understood tonogenetic process of segmental contrast neutralization. There are cases such as Swedish (Riad 1998), where a prominence marking pitch accent has been implicated in the development of lexical contrasts. However, the resulting contrasts are often realized in terms of early vs. late alignment. As such, there is no clear model through which pitch-accent reanalysis could develop into a contrast between H and L. I argue that Northern Pomo's reanalysis is driven by the differential effects of vowels and consonants on pitch perception which could lead a rising-falling accent to sound high on a short vowel and low on long vowels. After the association of a H tone to CV(C) and a L tone to CVV(C) is established, subsequent sound changes create new long vowels and shorten old long vowels, creating minimal pairs and rendering the relationship between phonemic vowel length and tone unpredictable - prompting these tones to become lexicalized.

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